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Aditya University

SCHOOL OF ENGINEERING

Basic Electrical and Electronics Engineering

Unit IV

SEMICONDUCTOR DEVICES

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Course Contents

UNIT – IV

SEMICONDUCTOR DEVICES :

Intrinsic semiconductors – Extrinsic semiconductors - P type and N type - P-N junction characteristics of P N Junction Diode — Zener Effect — Zener Diode and its Characteristics. working of simple zener voltage regulator and amplifier- Bipolar Junction Transistor — CB, CE, CC Configurations and Characteristics.

INTRODUCTION

- Electronics is the field of science and engineering that deals with the study, design and utilization of electron devices.
- The tremendous development in this field during last few decades makes us to live in an electronics world. Electronics plays a major role in almost every part of the life.
- The electronic appliances are used in homes, in offices and in industries. Some of the major application areas are listed below:

(i) Communication Systems

- **Telecommunications:** Electronics enable voice and data transmission over long distances through wired and wireless networks, including telephone systems, the internet, and cellular networks.
- **Satellite Communication:** Electronics are used in satellite systems for broadcasting, navigation, and global communication.

INTRODUCTION

(ii) Consumer Electronics:

- Smartphones
- Television and Audio Equipment
- Gaming Consoles
- Household Appliances

(iii) Computing and Information Technology:

- Personal Computers
- Servers and Data Centres
- Laptops and Tablets

INTRODUCTION

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- Medical Devices
- Health Monitoring

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- Avionics(aircraft systems)
- Defence Systems

INTRODUCTION

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(vii) Renewable Energy

- Solar Power
- Wind Power
- Energy storage

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(ix) Research and Scientific Instruments

(x) Telemedicine

(xi) Space Exploration

EVOLUTION OF ELECTRONICS - VACUUM TUBES TO NANO-ELECTRONICS

The evolution of electronics from vacuum tubes to nano-electronics is a fascinating journey marked by significant advancements in technology and miniaturization. Here's a summary of this evolution:

(i) **Vacuum Tubes (1900s-1950s):** Vacuum tubes, also known as electron tubes or thermionic valves, were the first practical electronic components.

(ii) **Transistors (1947):** The invention of the transistor at Bell Labs in 1947 by John Bardeen, Walter Brattain, and William Shockley revolutionized electronics Transistors were much smaller and more reliable than vacuum tubes. Transistors played a crucial role in the digital revolution, enabling smaller and more efficient electronic devices.

EVOLUTION OF ELECTRONICS - VACUUM TUBES TO NANO-ELECTRONICS

(iii) Integrated Circuits (ICs) (1950s-1960s): Jack Kilby and Robert Noyce independently developed the integrated circuit (IC) in the late 1950s and early 1960s. ICs allowed the integration of multiple transistors and other electronic components onto a single silicon chip. ICs paved the way for the development of powerful computers and a wide range of electronic devices.

(iv) Microelectronics (1970s-1980s): The 1970s and 1980s saw advancements in semiconductor manufacturing processes, enabling smaller and more densely packed transistors on ICs. Microelectronics played a significant role in the rise of personal computers.

EVOLUTION OF ELECTRONICS - VACUUM TUBES TO NANO-ELECTRONICS

(v) Nano-electronics (Late 20th Century Present): Nano-electronics focuses on electronic components at the nanoscale, typically involving structures with dimensions less than 100 nano-meters. Moore's Law continued to drive the miniaturization of transistors and ICs, predicting the doubling of transistor density roughly every 18 to 24 months. New materials and fabrication techniques, such as strained silicon and high-k dielectrics, have been explored to continue shrinking transistors.

(vi) Beyond Silicon (Ongoing): As transistors approached atomic scales, quantum mechanical effects began to play a significant role in their behavior. Quantum dot- based and spin-based (spintronics) devices have evolved as alternatives to traditional transistors. Quantum computers, which harness quantum properties to perform complex calculations, represent a revolutionary shift in computing technology.

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EVOLUTION OF ELECTRONICS - VACUUM TUBES TO NANO-ELECTRONICS

- The evolution from vacuum tubes to nano-electronics represents a remarkable progression technology, with each stage bringing about smaller, more efficient, and powerful electronic devices.
- Nano-electronics, in particular, opens the door to new possibilities in computing and electronic applications, pushing the boundaries of what is possible in the world of electronics.
- The developments in Electronics field are due to the revolution and innovation in **semiconductor materials and devices**.

SEMICONDUCTOR MATERIALS

- The structure of an atom is made up of three basic particles: protons, neutrons, and electrons.
- **Nucleus:** The nucleus is a tiny, positively charged center of the atom that contains protons and neutrons. The nucleus is less than one ten-thousandth the size of the atom, but contains more than 99.9% of its mass.
- **Protons:** Protons have a positive charge that is equal in magnitude to the negative charge of an electron.
- **Neutrons:** Neutrons have almost the same mass as protons, but have no charge.
- **Electrons:** Electrons are negatively charged and orbit the nucleus in shells. The outermost shell is called the valence shell.

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Energy Band Theory in Solids

- Band theory was developed by Felix Bloch in 1928.
- According to this theory, free electrons exhibit wave character as they move between atoms in a solid.
- This theory assumed that potential varies in a periodic manner in the solid. Free electrons move in periodic potential provided by lattice.
- This theory successfully explained the classification of solids into three groups, namely conductors, insulators and semiconductors.
- Based on band theory many of the electrical and thermal properties of solids could be successfully explained. It also formed the foundation of the technology of solid-state electronics.

VALENCE BAND, CONDUCTION BAND AND ENERGY GAP

VALENCE BAND(E_V) :

❖The Energy band occupied by valence electrons that are involved in covalent bonding, is called as valence band.

❖Depending upon the number of valence electrons this band may get partially or completely filled.

❖At absolute zero, covalent bonds are complete, therefore valence band is completely filled.

CONDUCTION BAND (E_c) :

❖The energy band above the valence band, having free electrons responsible for electrical conduction, is called as conduction band.

❖At absolute zero, this energy band is empty.

VALENCE BAND, CONDUCTION BAND AND ENERGY GAP

ENERGY GAP or FORBIDDEN ENERGY GAP (E_G):

- The energy interval between top of the valence band and bottom of the conduction band which is empty and forbidden is called energy gap or band gap.
- It is the special characteristic of semiconductor material.
- It is the minimum amount of energy required for breaking a covalent bond and to excite an electron from valence band to conduction band.

CLASSIFICATION OF SOLIDS BASED ON BAND THEORY

OR

Classification of Materials Based on Band Gap

❖Solids can be classified into conductors, semiconductors or insulators depending upon width of Energy gap.

❖Completely filled bands contain large number of electrons but do not contribute to the conductivity of the material.

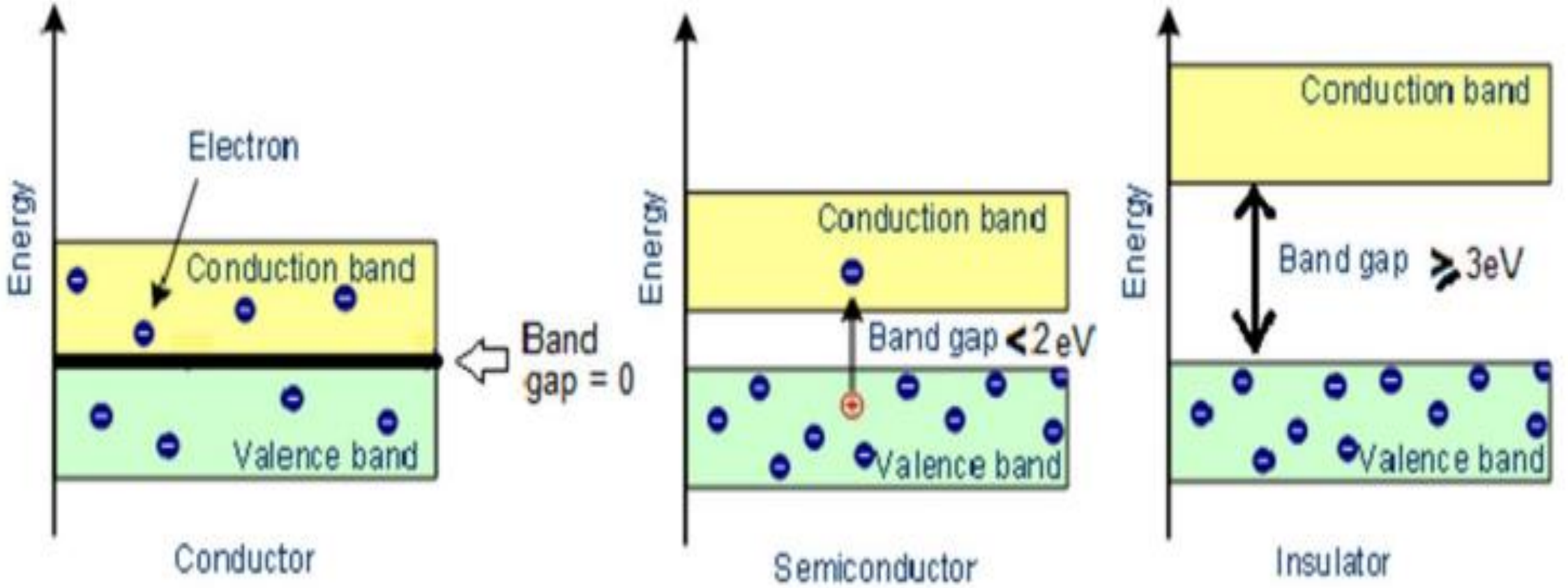
❖Partially filled bands are necessary for electrical conduction.

❖The energy band diagram of conductors, semiconductors or insulators is shown in figure

CLASSIFICATION OF SOLIDS BASED ON BAND THEORY

OR

Classification of Materials Based on Band Gap



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CLASSIFICATION OF SOLIDS BASED ON BAND THEORY

OR

Classification of Materials Based on Band Gap

Conductors:

❖The solids in which conduction and valence band overlap each other are called conductors. Therefore, the energy gap between valence band and conduction band is zero.

❖Electrons can easily jump from lower energy band to higher one and become available for conduction.

❖An application of a small amount of voltage leads to generation of large amount of current.

❖Hence these solids are good electrical conductors. For e.g., Lithium, Beryllium and sodium.

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CLASSIFICATION OF SOLIDS BASED ON BAND THEORY

OR

Classification of Materials Based on Band Gap

Semiconductors:

- The solids in which the conduction and valence bands are separated by a small energy gap of less than 1 eV or 2eV are called semiconductors.
- For e.g. Semiconductors like Silicon has Bandgap of 1.12 eV and Germanium has bandgap of 0.72 eV.
- A small energy gap means that a small amount of energy is required to free the electrons and move them from the valence band to the conduction band.
- The semiconductors behave like insulators at 0^0K, because valence electrons do not have required energy to jump to the conduction band.
- If the temperature is increased, valence electrons acquire sufficient energy to jump into the conduction band.
- Therefore, the conductivity of semiconductors increases with the increase in temperature

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CLASSIFICATION OF SOLIDS BASED ON BAND THEORY

OR

Classification of Materials Based on Band Gap

Insulators:

- The solids in which the conduction band and valence bands are separated by a large energy gap of ≥3 to 7 eV are called insulators.
- At room temperature, the valence electrons do not have enough energy to jump into the conduction band, therefore insulator do not conduct current.
- Thus, insulators have very high resistivity and extremely low conductivity at room temperatures. For e.g. Diamond and glass.

CLASSIFICATION OF SEMICONDUCTORS

Semiconductors can be broadly classified as

(i) Intrinsic (Pure) semiconductors and

(ii) Extrinsic (Doped) semiconductors

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INTRINSIC SEMICONDUCTORS

- Chemically pure semiconductors in which electrical conduction is due to thermally excited electrons and holes are known as intrinsic semiconductors. For eg. Germanium and silicon.
- In these semiconductors, conductivity is mainly due to electron-hole pairs generated due to thermal agitation.
- The electrons reaching the conduction band, due to thermal excitation leave equal number of holes in valence band.
- Hence number of free electrons in conduction band is always equal to the number of holes in the valence band.

ENERGY LEVEL DIAGRAM OF INTRINSIC SEMICONDUCTOR

At T=0K

- At 0K, all valence electrons of each silicon atom are involved in covalent bond with neighboring atoms and has no free electron is available to participate in the process of conduction.
- Under these conditions, valence band is completely filled and conduction band is completely empty as shown in Fig.1.10.
- Electrons in the valence band cannot cross the forbidden energy gap and hence the material behaves as an insulator.

ENERGY LEVEL DIAGRAM OF INTRINSIC SEMICONDUCTOR

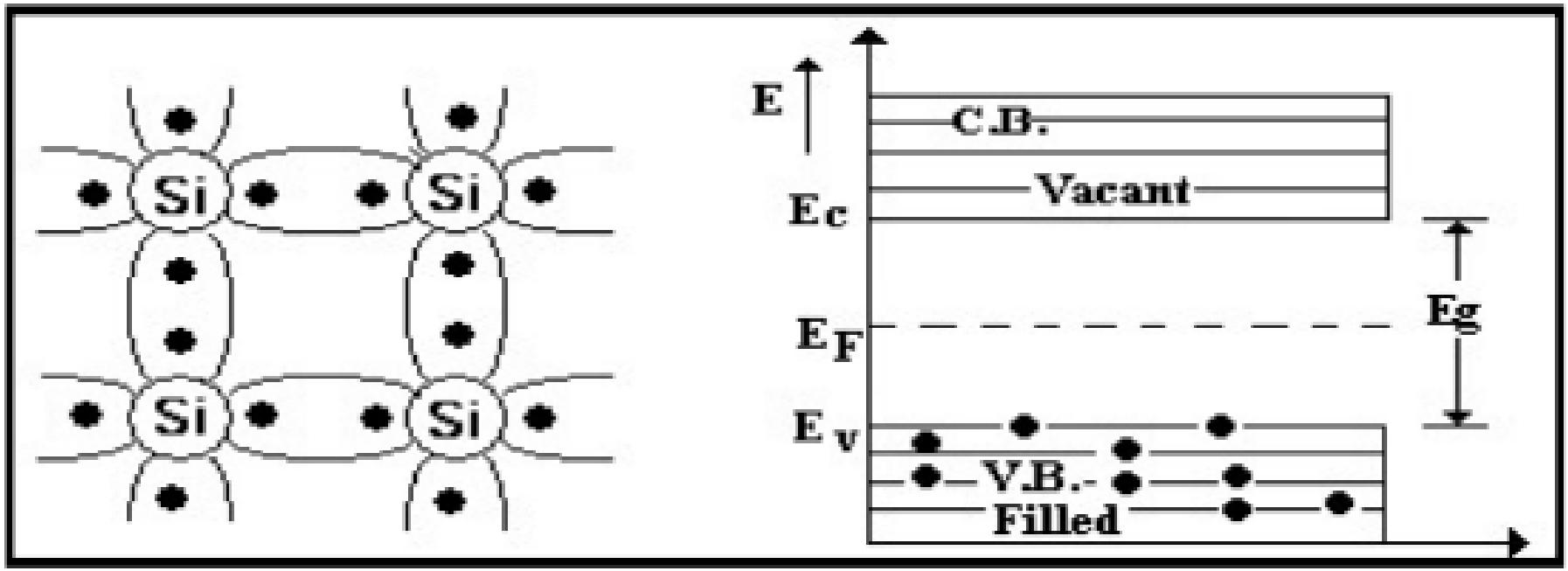


Fig. 1.10: (a) 2D representation of silicon crystal (b) Energy band diagram of silicon at T = 0K

ENERGY LEVEL DIAGRAM OF INTRINSIC SEMICONDUCTOR

AT T>0K

- As temperature increases, electrons acquire sufficient energy to break the covalent bonds due to the supply of thermal energy and jumps from valance band to conduction band after gaining energy equal to Eg.
- Whenever one electron is made available for conduction, a hole is formed in valance band at the same time as shown in Fig.1.11.
- If n= number of electrons, p= number of holes and ni= intrinsic carrier concentration or density. Hence, n = p = ni,
- As the concentration of electrons in conduction band and valence band are equal, Fermi level E_F is located exactly at the middle of forbidden gap (Eg).

ENERGY LEVEL DIAGRAM OF INTRINSIC SEMICONDUCTOR

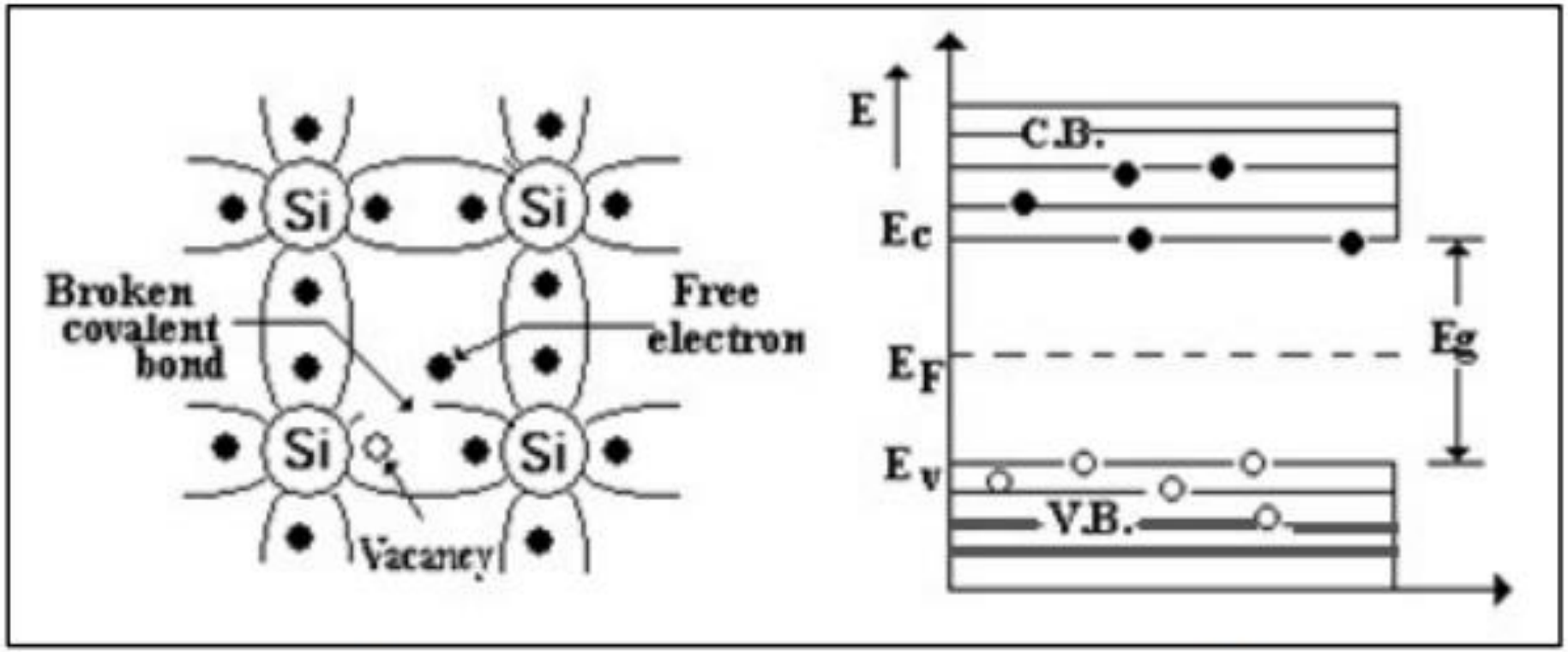


Fig. 1.11: (a) 2D representation of silicon crystal (b) Energy band diagram of silicon at T > 0K.

EXTRINSIC SEMICONDUCTORS

- The electronic properties and the conductivity of an intrinsic semiconductor can be changed in a controlled manner by adding very small quantities of other elements called dopants.
- This can be achieved by adding impurities of III or IV group elements to the melt and then allowing it to solidify into the crystal.
- This process is called doping and these doped semiconductors are called extrinsic semiconductors.

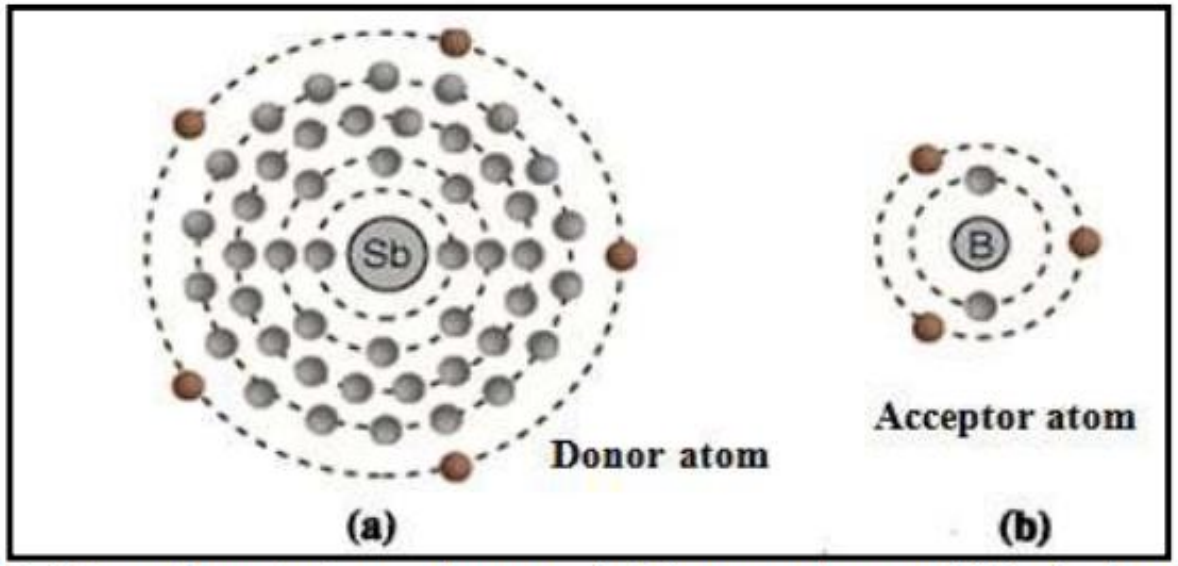


Fig.1.12: Illustration of valence electrons in (a) pentavalent and (b) trivalent impurities.

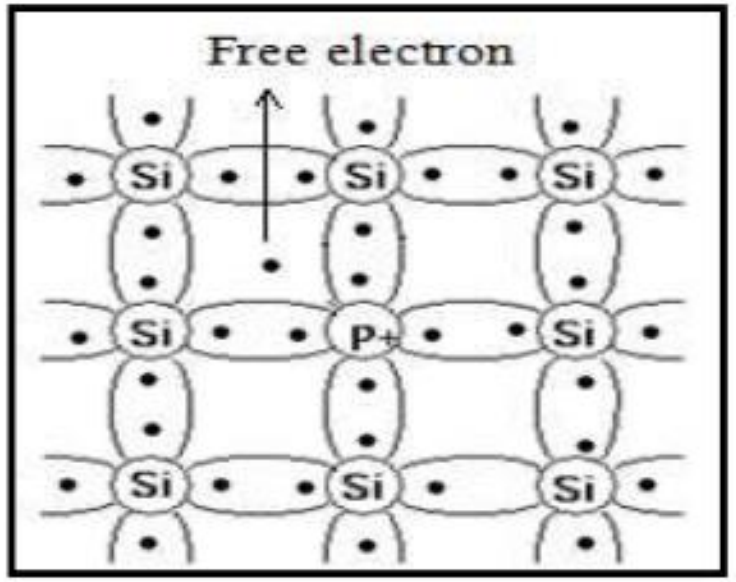
EXTRINSIC SEMICONDUCTORS

Doping is done at a rate such that only one atom of impurity is added per 106 to 1010 semiconductor atoms. The impurities are the materials used to dope the intrinsic semiconductor materials. These materials can be of two types, such as (i) donor impurity and (ii) acceptor impurity. Extrinsic Semiconductors are further classified into two categories:

- N-type semiconductor → Pure semiconductor + group V element [pentavalent impurity]
- P-type semiconductor → Pure semiconductor + group III element [trivalent impurity]
- Impurity atoms with 5 valence electrons produce N-type semiconductors by contributing extra electrons,
- Impurity atoms with 3 valence electrons produce P-type semiconductors by producing a "hole" or electron deficiency.

N-TYPE SEMICONDUCTOR

- An n-type semiconductor is produced when a pure semiconductor is doped with a pentavalent impurity (having 5-electrons in the outermost orbit) such as Phosphorus, Arsenic or Antimony. The impurity atom has one excess unpaired electron as shown in figure1.13. At room temperature thermal energy is sufficient to make this electron free. This impurity is called donor impurity, as it donates electrons. The addition of donor impurity increases the number of electrons in the semiconductor. Hence in N type semiconductor, electrons are majority charge carriers and holes are minority charge carriers.



ENERGY BAND DIAGRAM OF N-TYPE SEMICONDUCTOR

- Figure 1.14 shows the Energy band diagram of N-type semiconductor.
- At 0K, the donor level is located below the bottom of conduction band. The donor level is filled by donor atoms and fermi level EF is located between the bottom of conduction band and donor level.
- Thus, at 0K, the donor atoms are not ionized, the conduction band is empty while the valence band is full and the material behaves as an insulator(Fig.1.14a).
- While at higher temperature (T > 0K), the electrons from the donor level and from the valence band jump to conduction band and material shows the conductivity(Fig.1.14 b).
- Conduction band contains large number of electrons donated by impurity atoms in addition to electrons jumping from valence band.

ENERGY BAND DIAGRAM OF N-TYPE SEMICONDUCTOR

- The impurity atoms become positive ions (they are ionized). They gain a positive charge due to donation of electrons. The donor level is now empty as it is devoid of electrons. It contains only positive ions. The fermi level E_F shifts below the donor level as shown in fig.1.14(b).

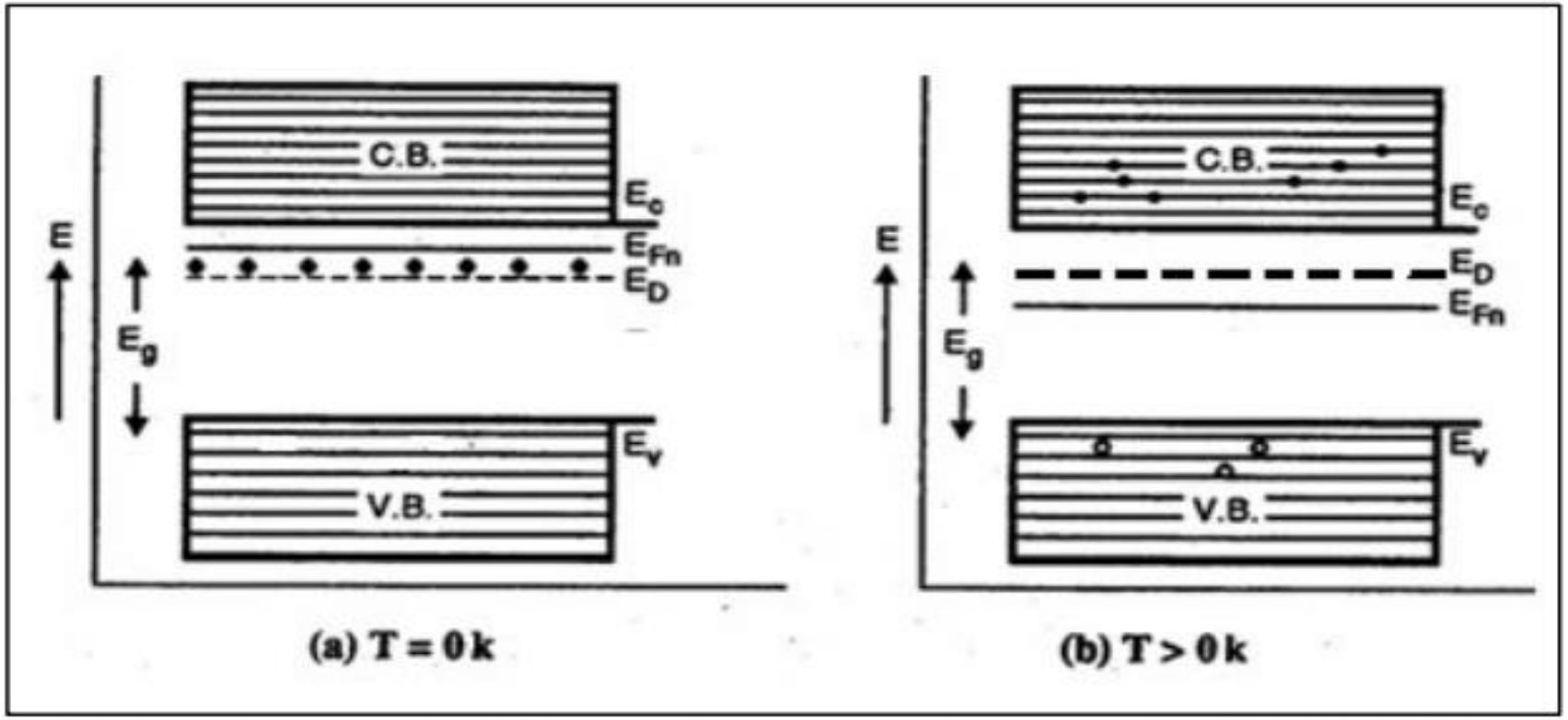


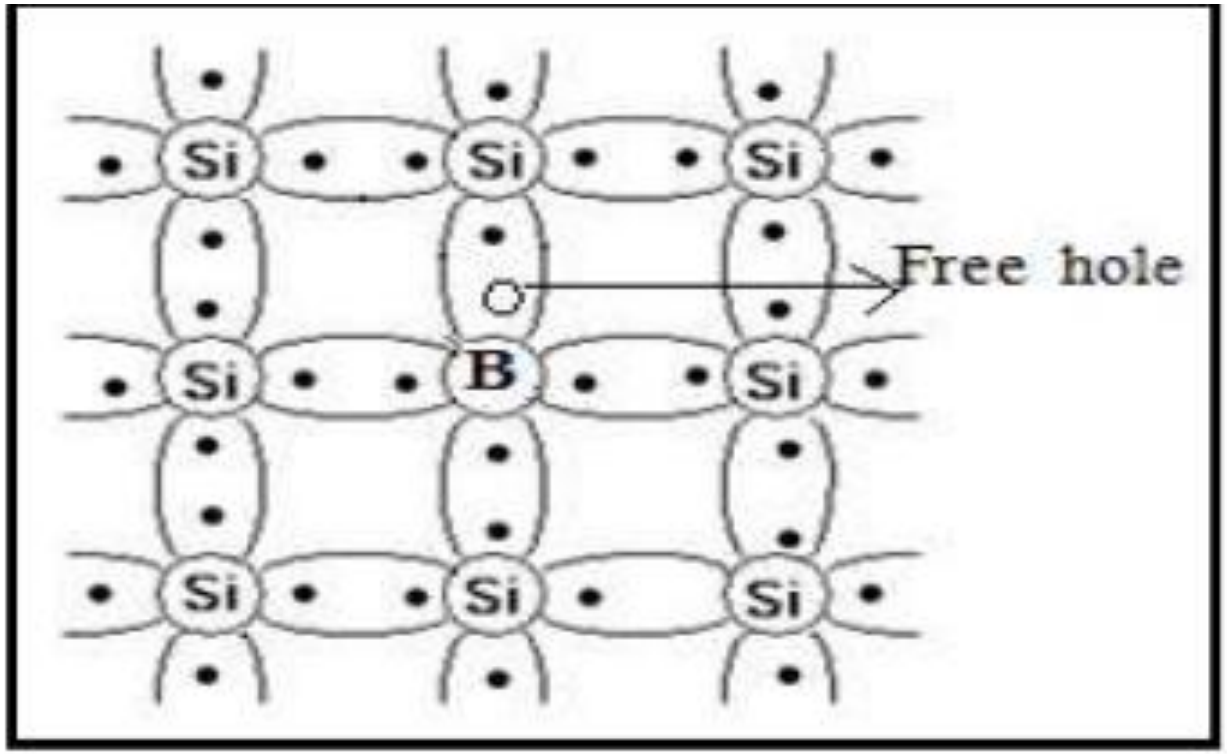
Fig.1.14: Energy band diagram of N-type semiconductor at (a) At 0K (b) At T > 0K.

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P-TYPE SEMICONDUCTOR

- A p-type semiconductor is produced when a pure semiconductor is doped with a trivalent impurity (three electrons in outer most orbit) such as Boron, Aluminium, Gallium or Indium.
- Them impurity atom is deficient of one electron. The absence of one electron is treated as hole as shown in fig.1.15.
- The impurity is called acceptor impurity, as it accepts one electron from neighboring semiconductor atoms if small amount of energy is supplied.
- Then hole will be created in the neighboring semiconductor atom. The impurity atom supplied holes which are ready to accept electrons.
- The number of holes is thus more than electrons in P-type semiconductor. Therefore, holes are majority charge carriers and electrons are minority charge carriers in P-type semiconductor. The conductivity is mainly due to holes in this semiconductor.

P-TYPE SEMICONDUCTOR



ENERGY BAND DIAGRAM OF P-TYPE SEMICONDUCTOR

- The energy band diagram of P-type semiconductor is shown in fig.1.16. At 0 K, the acceptor levels are devoid of electrons and are hence empty. The valence band is full and conduction band is empty, the material behaves as an insulator (Fig.1.16a).
- The acceptor level is located above the valence band and fermi level EF is between acceptor level and top of valence band. While at higher temperature (T > 0K), the electrons from valence band jump to acceptor level and also into the conduction band leaving holes in valence band, hence material shows the conductivity (Fig.1.16b).
- The acceptor level is now filled with electrons. It contains ionized atoms carrying negative charge. The fermi level E_F shifts above the acceptor level. In P type semiconductor, holes are majority charge carriers and electrons are minority charge carriers.

ENERGY BAND DIAGRAM OF P-TYPE SEMICONDUCTOR

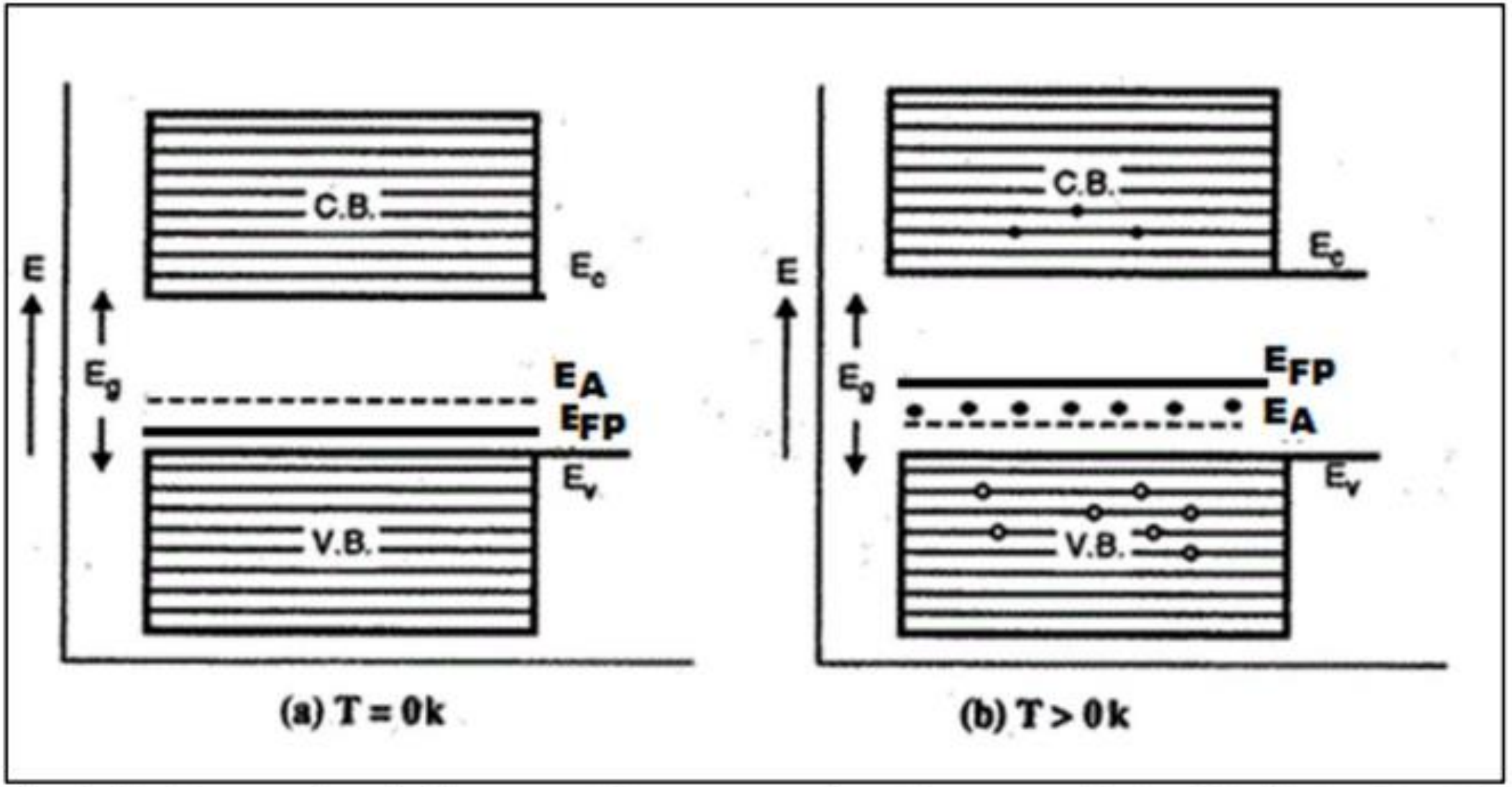


Fig.1.16: Energy band diagram of P-type semiconductor at (a) At 0K (b) At T > 0K

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DIFFERENCE BETWEEN INTRINSIC & EXTRINSIC

	Intrinsic Semiconductors	Extrinsic Semiconductors
Doping	No intentional doping with impurity atoms.	Intentional introduction of impurity atoms through doping.
Type of Atoms	Made of a single type of atom (e.g., Silicon or Germanium).	Doped with specific impurity atoms — **P-type** — Trivalent **N-type** — Pentavalent
Charge Carriers	Nearly equal numbers of electrons and holes.	The type of doping determines dominant charge carriers.
Conductivity Type	Exhibits both n-type and p-type conductivity.	Exhibits n-type or p-type conductivity based on doping.
Majority Charge Carriers	Electrons and holes have similar concentrations.	Either electrons (n-type) or holes (p-type) can be the majority carriers.
Electrical Conductivity	Relatively low electrical conductivity.	Enhanced electrical conductivity due to intentional doping.
Energy Levels	Valence and conduction bands remain relatively close.	Energy levels are modified, creating a wider bandgap (doping).
Bandgap Modification	The bandgap remains relatively constant.	Bandgap can be modified based on the type and concentration of dopants.
Applications	Not commonly used for practical devices.	Used extensively in electronic devices like diodes or transistors.
Examples	Silicon (Si), Germanium (Ge)	**N-type**: Phosphorus (P), Antimony (Sb) **P-type**: Boron (B), Indium (In)

DIFFERENCE BETWEEN N-TYPE and P-TYPE SEMICONDUCTOR

n-type semiconductor	**p-type semiconductor**
1. Electrons are majority charge carriers and holes are minority charge carriers.	1. Holes are majority charge carriers and electrons are minority charge carriers.
2. The majority of charge carriers move from low potential to high potential.	2. The majority of charge carriers move from high potential to low potential.
3. An donor is a doping atom that can produce an n-type semiconductor whenever introduced to a semiconductor.	3. An acceptor is a contaminating atom that could produce a p-type semiconductor whenever introduced to a semiconductor.
4. Pentavalent impurities are added.	4. Trivalent impurities are added.
5. It has donor energy levels that are very close to the conduction band.	5. It has acceptor energy levels that are very close to the valence band.

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P-N junction

- The PN junction diode is a semiconductor device with two semiconductor materials in physical contact, one with excess of holes (P-type) and other with excess of electrons (N-type).
- A PN junction diode may be formed from a single-crystal intrinsic semiconductor by doping part of it with acceptor impurities and the remaining with donors. Such junctions can form the basis of very efficient rectifiers.
- The most important characteristic of a PN junction is its ability to allow the flow of current in only one direction. In the opposite direction, it offers very high resistance. The high vacuum diode has largely been replaced by silicon and selenium rectifiers.
- Semiconductor diodes find wide applications in all phases of electronics, viz. radio and TV, optoelectronics, power supplies, industrial electronics, instrumentation, computers, etc.

P-N junction Diode

- A semiconductor PN junction is formed when a wafer of semiconductor material such silicon, if one half is doped by p-type impurity and other half is doped by n-type impurity as shown in figure (6.8), The n-type material has high concentration of electrons and p material has more number of holes. Due to the variations in charge a tendency for the free electrons to diffuse over to the p-side and holes to the n-side. This process is called diffusion.

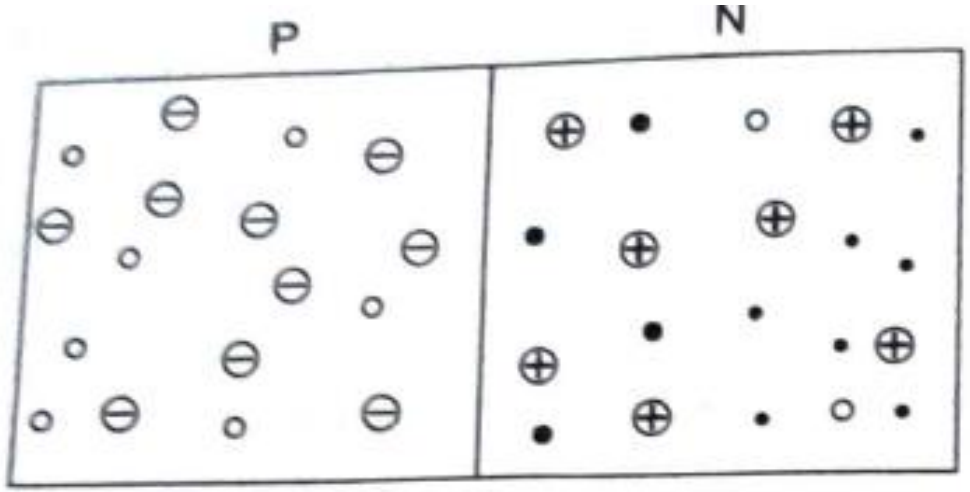


Figure (6.8): *PN Junction*

P-N junction Diode

- The process of diffusion takes place only for a short period. As the free electrons move across the junction from n-type to p-type, the donor ions become positively charged. Hence a positive charge is built on the n-side of the junction. The free electrons that cross the junction recombine with holes leaving the negative acceptor ions uncovered. The process of recombining an electron with a hole is called recombination. As a result, a net negative charge is established on the P-side of the junction as shown in figure (6.9).

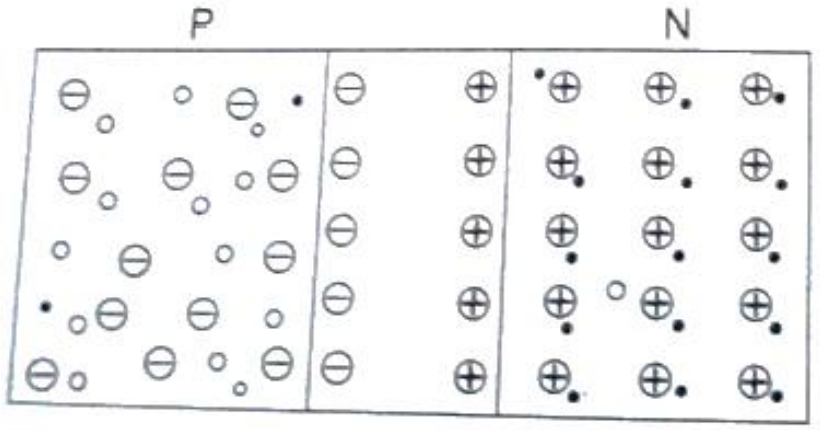


Figure (6.9): *Formation of depletion region in PN Junction*

• → Electron, ∘ → Hole, ⊖ → Acceptor ion, ⊕ → Donor ion

P-N junction Diode

- A restraining force is being set up in the neighbourhood of the junction. Thus a barrier is created near the junction which prevents further movement of charge carriers i.e. electrons into P-side and holes into N-type. The presence of immobile positive and negative ions on opposite sides of the junction, an electric field is induced across the junction. This electric field is known as the barrier potential or junction potential. The barrier potential represents the height of the barrier that is to be overcome by the electrons and holes to cross the junction.
- The barrier potential or contact potential for silicon, it is about 0.7 V and for germanium, it is about 0.3 V. The region having the uncovered acceptor and donor ions is called depletion region. The reason for this is that there is a separation of mobile charge carriers in this region .this region contain immobile or fixed ions which are electrically charged. Therefore ,this depletion region is also called as space-charge region.

Operation of PN Junction

- when the PN Junction is formed, the depletion region (space-charge region) is created and due to this electrons and holes cannot cross the barrier. Thus,the current flowing through an unbaised P-N junction is zero .
- When an external dc source is applied across the PN Junction is called as biasing of PN Junction. By biasing the PN junction, the width of the depletion region can be controlled. The PN junction can be biased in two ways, such as (i) forward bias, and (ii) reverse bias.

Operation of PN Junction

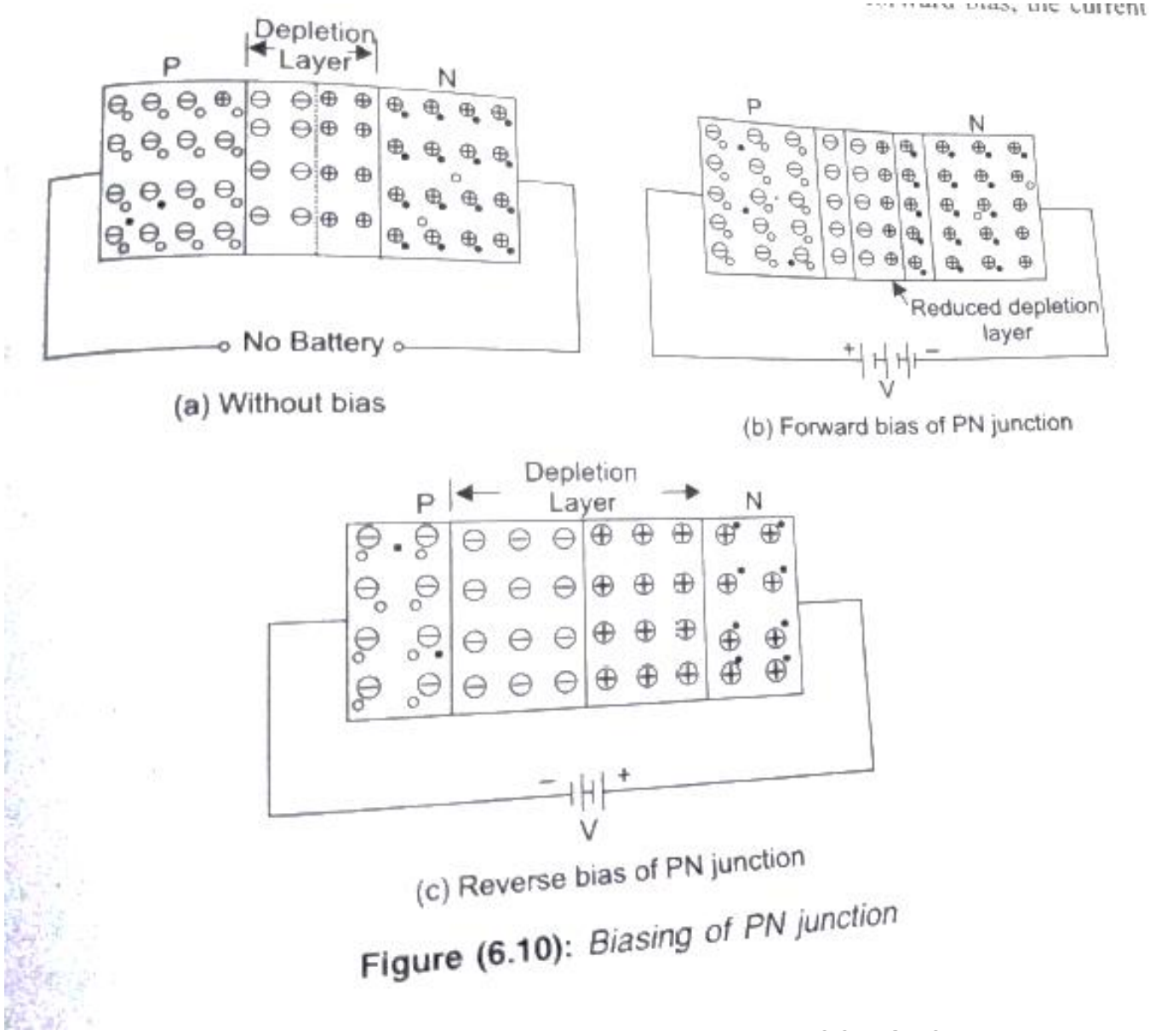


Figure (6.10): *Biasing of PN junction*

Operation of PN Junction-Forward Bias

- In the forward bias of a PN junction, positive terminal of the battery is connected to the p-side and the negative terminal to the n-side as shown in figure 6.10(b). By forward biasing, electrons in the N region are repelled from the negative terminal of the battery and forced towards the junction. Similarly, holes in the p region are repelled from the positive terminal of the battery and forced towards the junction.
- Due to this external energy, some holes and electrons enter the depletion region. This makes the uncovered charges to be mobile. This reduces the potential barrier as well the depletion region width. Consequently, more majority carriers diffuse across the junction. Hence, this causes a large current to flow across the junction. As long as the external battery is connected the PN junction, large amount of forward current is possible. In forward bias, the current due to majority carriers.

Operation of PN Junction-Reverse Bias

- In the reverse-bias of a PN junction, a battery connected across the PN junction such that its positive terminal is connected to the n-region and negative terminal is connected to the p-region as shown in figure (6.10c) .In reverse bias the electrons in the n-region are attracted towards the positive terminal of the battery whereas the holes in the p-region are being attracted towards the negative terminal of the battery. So in this process the majority charge carriers are drowning away from the PN junction. Due to this, the depletion region width increases, this in turn increases the barrier potential. As result, of the increased barrier potential, the majority charge carriers are not able to cross the junction. Therefore, in reverse bias, there is no current due to the majority charge carriers. But, there are few thermally generated minority carriers in both the regions.

Operation of PN Junction-Reverse Bias

- The increased barrier potential increases the flow of minority carriers across the junction. The generated minority carriers are drifted towards the PN-junction due to the barrier potential. As only the minority carriers are involved, a very small amount of current flows through the PN-junction.
- The generation of minority carriers depends upon the temperature and is independent of the reverse voltage applied. Therefore, if temperature is constant, the current due to the minority carriers also remains constant whether the reverse bias voltage is increased or decreased. This current is called as reverse saturation current. The value of reverse saturation current is of the order of μA for germanium PN-junction and is of the order of nA for silicon PN-junction,

PN Junction Diode

Biasing of diode	Terminal	Charge	Graphic Symbol
Forward	P-side (Anode)	Positive	giving +ve to anode and –ve to cathode 
	N-side (Cathode)	Negative	
Reverse	P-side (Anode)	Negative	giving -ve to anode and +ve to cathode 
	N-side (Cathode)	Positive	

The symbol for a diode is an arrowhead and a bar. The arrowhead indicates the conventional direction of current flow when the diode is forward biased, i.e. from positive terminal through the device to the negative terminal.

V-I Characteristics

Forward Characteristics of a PN Junction:

- A semiconductor diode's voltage-current, or V-I, characteristic curve is assigned to the behavior of the depletion layer, which exists at the P-N junction between the two semiconductors.
- In the forward bias condition, the depletion region is reduced in width and the barrier potential progressively disappears as the applied bias voltage is increased from zero. The current which flows is called 'Forward Current (I_F)'
- A very small amount of forward current flows until V, exceeds the junction barrier potential. The characteristics follow the exponential law.

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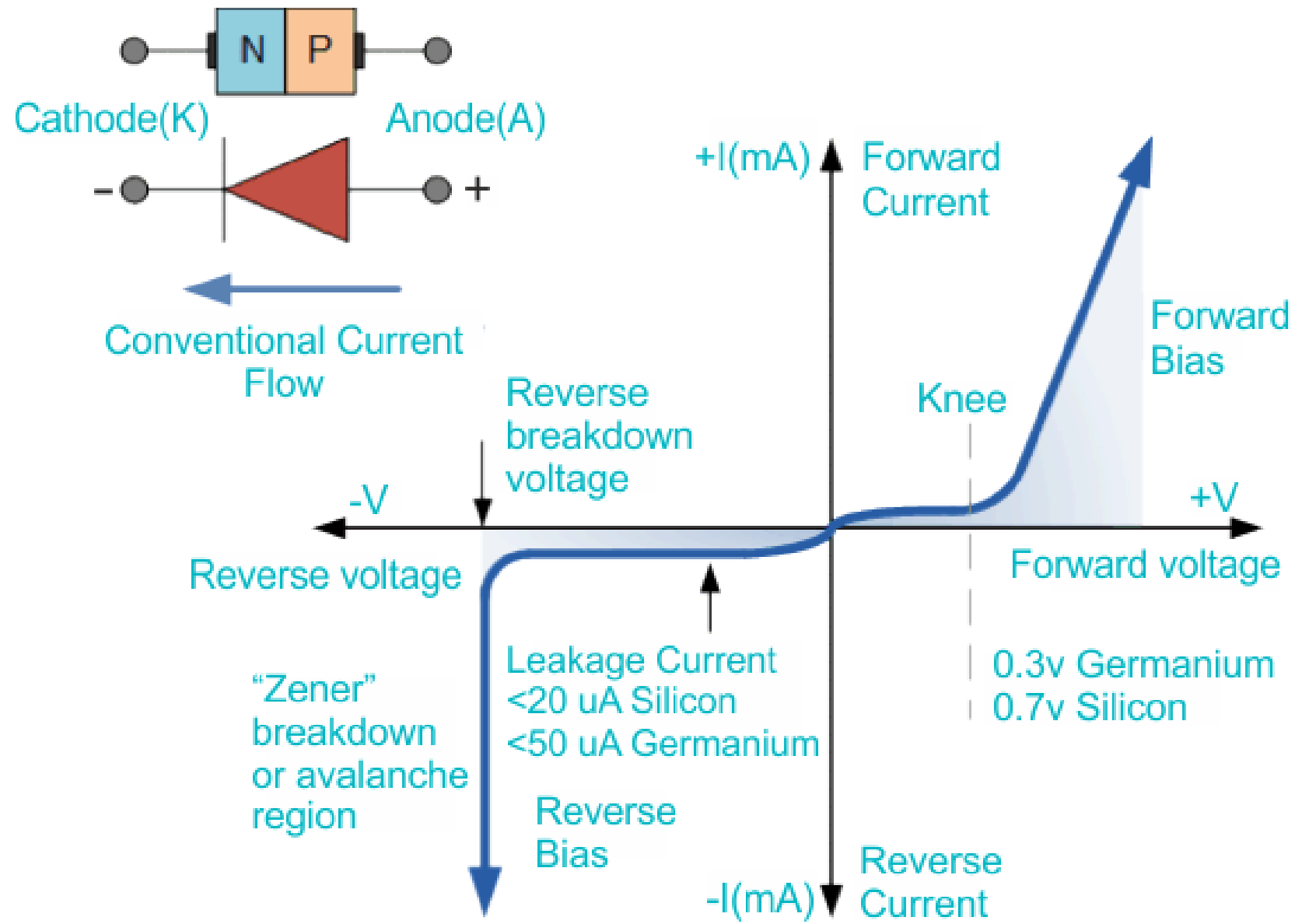
V-I Characteristics

Reverse Characteristics of a PN Junction:

- VI characteristics is the graph of reverse current I_R, versus reverse voltage V_R. When a PN junction is reverse biased, the depletion region is widened. There is no possibility of majority carrier current flow.
- However, minority carriers generated on each side can still cross the junction. This current is called as 'Reverse Current (I_R)'.
- When all the minority carriers are flowing across, further increase in bias voltage will not increase the current. This current is referred to as Reverse Saturation Current, denoted by I_R or I_o

V-I Characteristics

testbook



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Diode Equation

The diode equation describes the diode behaviour in the forward and reverse biased regions.

It gives the relationship between current through the junction diode and voltage across it.

$$I_D = I_S\left(e^{qV_D/NKT} - 1\right)$$

Where,

I_D = Diode current in amps

I_S = Saturation current in amps

(typically 1×10^{-12} amps)

e = Euler's constant (:2.718281828)

q = charge of electron (1.6×10^{-19} coulombs)

V_D = Voltage applied across diode in volts

N = "Nonideality" or "emission" coefficient

(typically between 1 and 2)

k = Boltzmann's constant (1.38×10^{-23})

T = Junction temperature in Kelvins

$$I_D = I_S\left(e^{V_D/0.026} - 1\right)$$

$$I = I_S e^{V_D/(nV_T)}$$

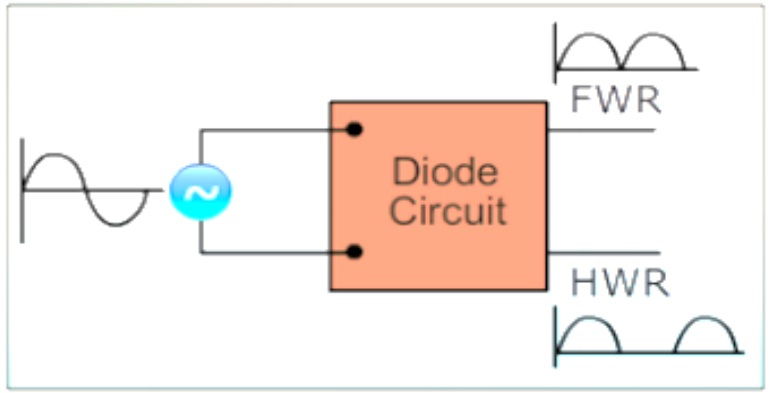
V_T = Thermal Voltage = 23mV

at room temp

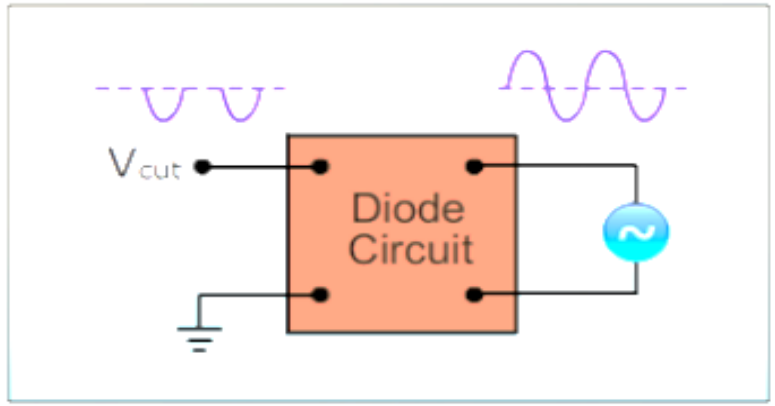
Applications

PN Junction diode has a vast range of applications

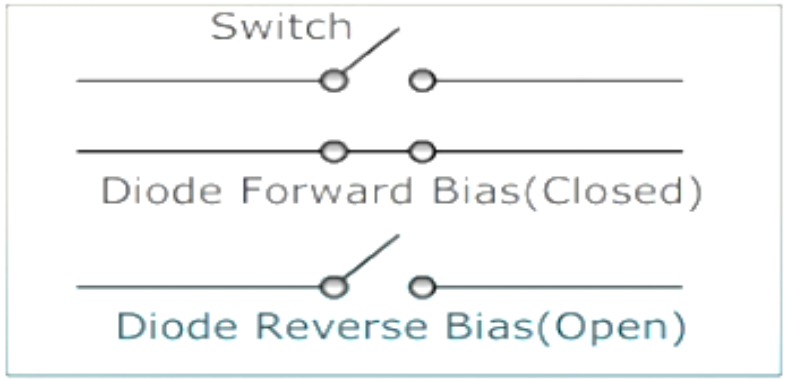
The typical applications include.



Rectifiers



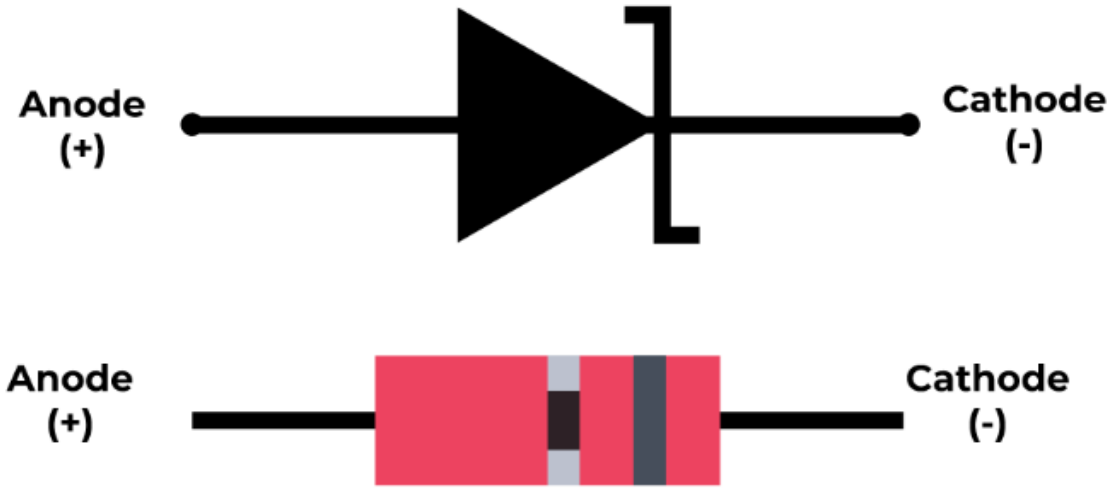
Clippers



Switch

Zener Diode

- Zener diode is a heavily doped pn junction diode operates in the breakdown region during reverse bias. The Zener diode allows current to flow from anode to its cathode like a normal semiconductor diode in forward bias. When the reverse voltage reaches breakdown voltage in an ordinary diode, the current through the junction and the power dissipated at the junction will be high. The ordinary diode gets damaged. The diodes can be designed to operate in the breakdown region by heavily doping such as Zener diode. The circuit symbol of Zener is shown in figure



Operating Principle

The Zener diode when it works in reverse bias provides a large amount of current at a breakdown voltage point. The electrical breakdown can happen in two approaches such as

(i) Zener breakdown, and

(ii) Avalanche breakdown.

i) Zener Breakdown: The Zener breakdown occurs in the Zener diodes having V_z less than 6V or less. In Zener diode, the P and N regions are heavily doped and direct rupture of covalent bonds takes place due to the strong electric fights, at the PN junction. The electrons, thus free are available for conduction. The increase in free electrons contributes to large reverse current at a constant value of reverse bias and breakdown is said to have occurred due to **Zener effect**.

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Operating Principle

(ii) Avalanche Breakdown: The Avalanche breakdown takes place in Zener diodes having V_z more than 8V. In the reverse bias, the conduction is due to the minority carriers. As the reverse bias increases, these minority carriers tend to accelerate. The kinetic energy associated with them increases. These accelerated electrons rupture the covalent bonds and create new electron hole pairs. The new carriers upon acquiring sufficient energy from the field collide with other immobile ions to generate further electron hole pairs. This phenomenon is cumulative in nature and the mechanism of carrier generation is known as **Avalanche multiplication**. A large amount of current flows at the same value of reverse bias due to the **Avalanche breakdown**.

V-I Characteristics of Zener Diode

The behavior of Zener diode is similar to that an ordinary diode in forward bias and the behavior in reverse bias is based on the breakdown phenomenon.

Forward bias of Zener diode: When the anode of the Zener diode is connected to the positive terminal of the de source and the cathode is connected to the negative terminal, then the Zener diode is said to be forward biased. The behavior of forward biased Zener diode is identical to the forward biased ordinary diode. The circuit of forward bias is shown in figure

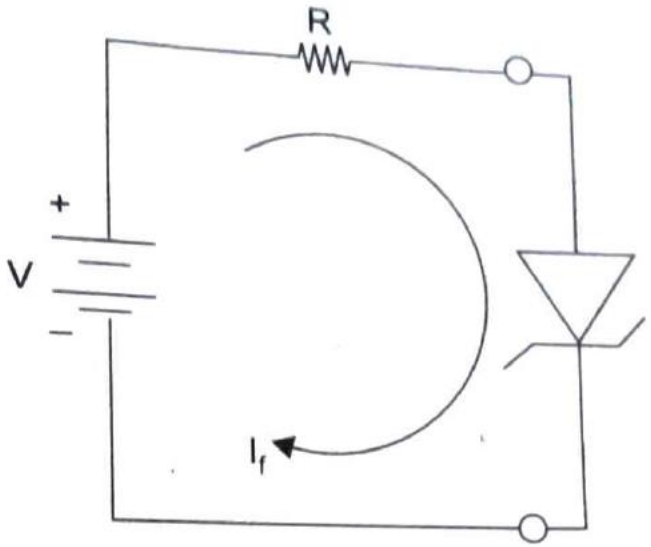


Figure (6.15): *Zener Diode in forward bias*

V-I Characteristics of Zener Diode

Reverse bias of Zener diode: When the cathode is connected to the positive terminal and anode is connected to the negative terminal of the de source, the Zener diode is said to be reverse biased. The circuit of reverse bias is shown in figure . In the reverse bias if the voltage applied is increased, the junction breaks down and reverse current increase sharply and is limited by the external resistance connected in series. The voltage at this point is called **breakdown voltage (V_z)**. The breakdown voltage depends on the width of the depletion layer. This width of the depletion layer depends upon the doping level. There are two mechanisms by which the Zener diode can breakdown, such as (i) Zener breakdown and (ii) Avalanche breakdown.

V-I Characteristics of Zener Diode

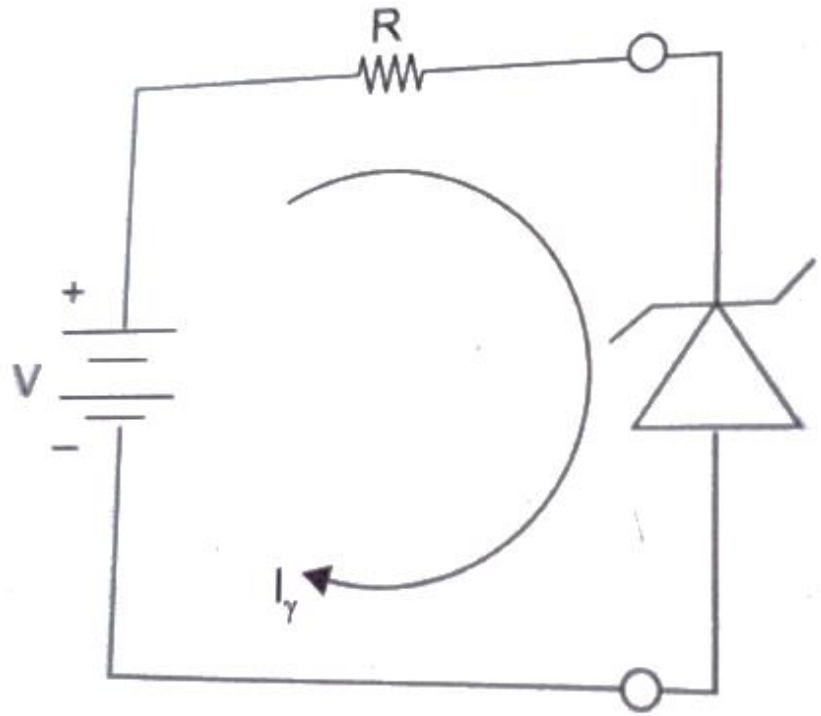
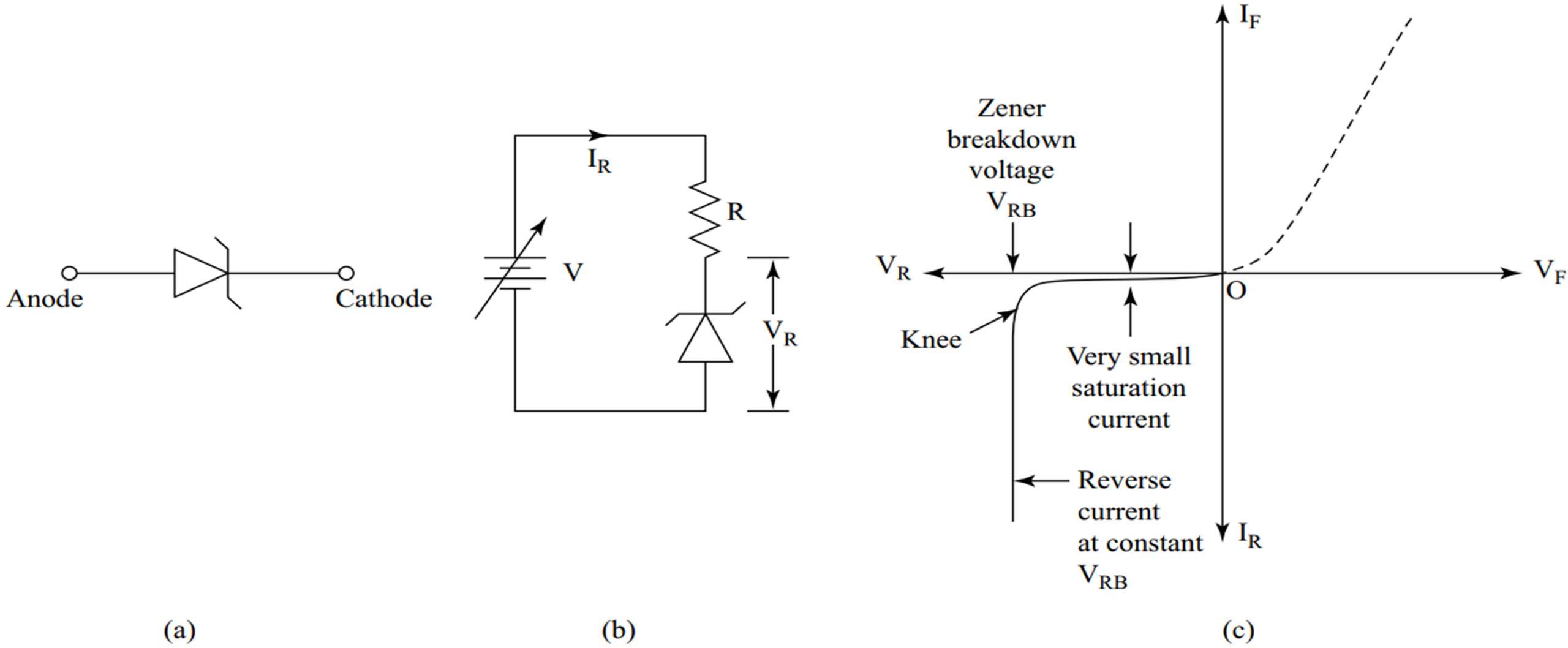


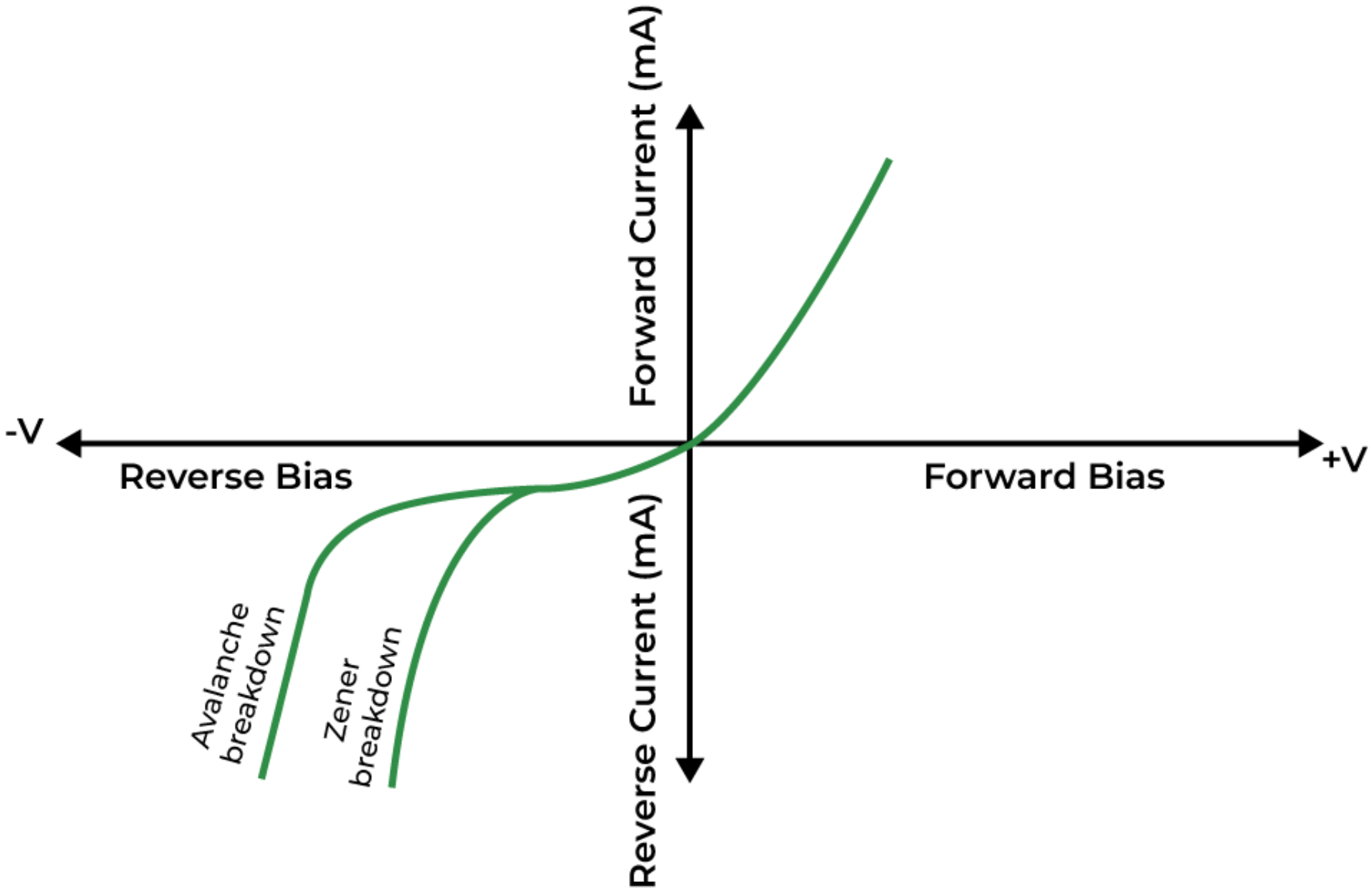
Figure (6.16): *Zener Diode in reverse bias*

The symbol, the circuit, and the V–I characteristic of a zener diode are shown in Fig.



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V-I Characteristics of Zener Diode



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Avalanche Breakdown vs Zener Breakdown

Avalanche Breakdown	Zener Breakdown
Avalanche breakdown occurs when the high voltage increase the free electron in the semiconductor and a sudden increase in current is seen.	Zener breakdown happens when electrons from the valance band gain energy and reaches the conduction band which then conducts electricity.
Avalanche breakdown is seen in the diodes having breakdown voltage greater than 8 volts.	Zener breakdown is seen in the diodes having breakdown voltage in the range of 5 to 8 volts.
Avalanche breakdown is observed in diodes that are lightly doped.	Zener breakdown is observed in diodes that are highly doped.
In the Avalanche breakdown, the VI characteristics curve is not as sharp as the VI characteristics curve in the Zener breakdown.	Zener Breakdown has a sharp VI characteristics curve.
For Avalanche breakdown increase in temperature increases the breakdown voltage.	For Zener breakdown increase in temperature decreases the breakdown voltage.

Applications of Zener Diode

Various applications of Zener diode are as listed below:

- Voltage regulator
- Overvoltage protection
- Used in Clipping and Clamping circuits
- Surge suppression circuitry for device protection
- Used in switching applications

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Difference between PN junction diode and Zener Diode

Parameter	PN Junction Diode	Zener Diode
Definition	A two-terminal semiconductor device formed by joining a p-type semiconductor with an n-type semiconductor by alloying is known as PN junction diode.	A Zener diode is a special type of two terminal semiconductor device which is optimized to work in breakdown region.
Doping level of p-type & n-type semiconductors	The level of doping of p-type and n-type semiconductors used for constructing PN junction diode is low.	The highly doped p-type and n-type semiconductors are used for making Zener diode.
Width of depletion layer	As the PN junction diode has low doping level, thus the width of its depletion layer is more.	The Zener diode is highly doped, hence its depletion region is narrow.
Effect of reverse current	The high reverse current can damage the PN junction diode.	The Zener diode does not get damaged by the reverse current.

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Difference between PN junction diode and Zener Diode

Parameter	PN Junction Diode	Zener Diode
Breakdown voltage	In case PN junction diode, the breakdown voltage is high.	The Zener diode has relatively lower breakdown voltage.
Biasing condition	The PN junction diode is designed to operate in forward bias only, i.e. current flows from anode to cathode.	The Zener diode is mainly designed to operate in reverse biased conditions.
Ohm's law	The PN junction obeys Ohm's law.	The Zener diode does not obey Ohm's law.
Manufacturing materials	The PN junction diode is generally made by silicon and germanium. Silicon is extensively used.	The material used for making the Zener diode is selected according to the required voltage.

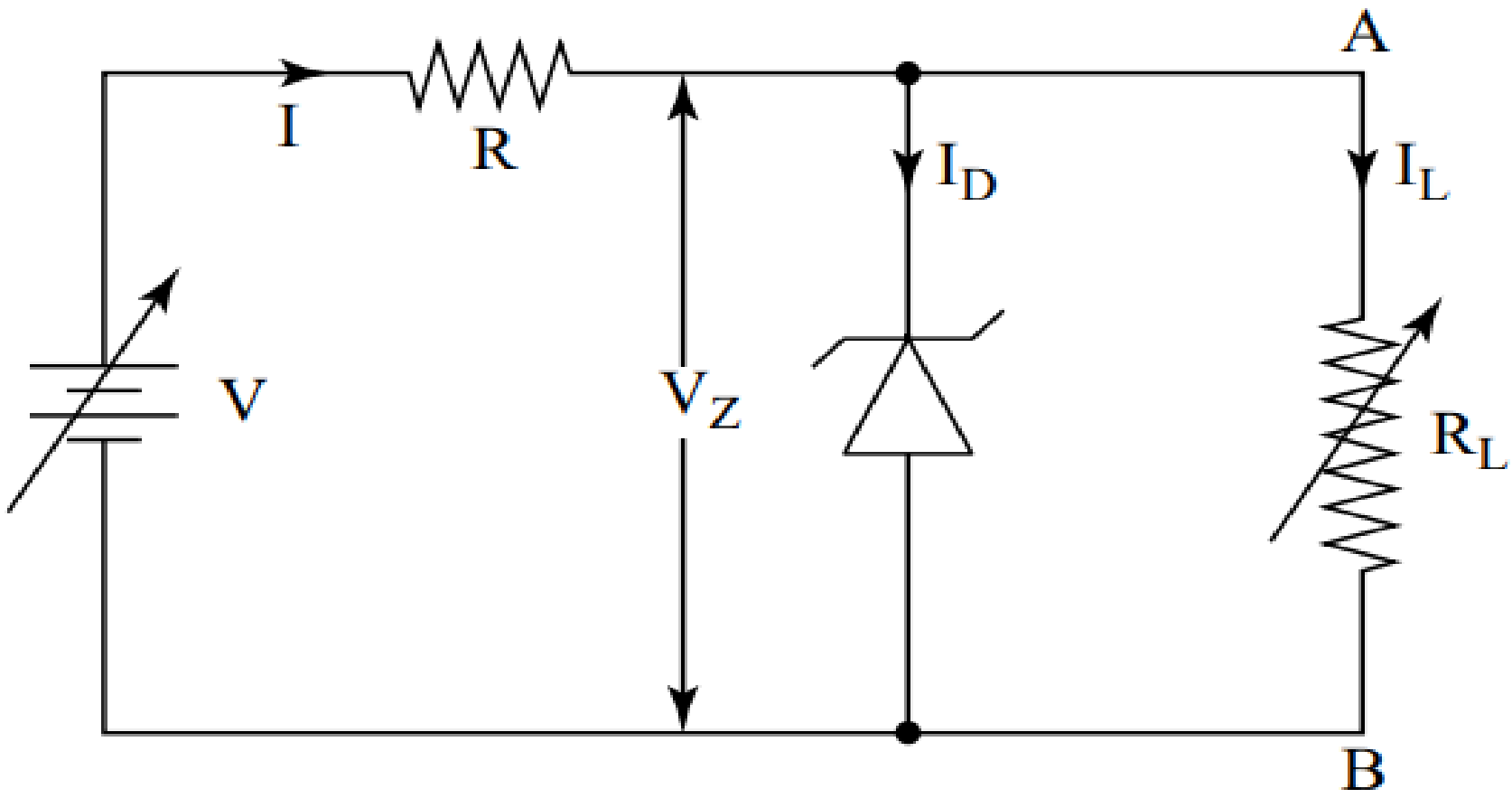
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Difference between PN junction diode and Zener Diode

Parameter	PN Junction Diode	Zener Diode
Current flow	In PN junction diode, the current flows only in one direction (or forward direction).	In case of Zener diode, current can flow in both directions.
Behave in forward & reverse bias	A forward biased ideal PN junction diode acts a closed switch while in the reverse bias acts as an open switch, i.e. do not conduct.	Zener diode acts as a normal PN junction diode in the forward bias condition, but it also conducts in the reverse bias condition.
Application	The PN junction diode extensively used in rectification, clamping, clipping, etc.	The Zener diode is mainly used as a voltage regulator (or stabilizer) in the electronic devices and circuits.

ZENER VOLTAGE REGULATOR

- A Voltage regulator maintains nearly constant voltage output across the load over a wide range of variation of load current.
- The Zener diode used in the circuit is shown in Fig. maintains a constant voltage across the load terminals A and B.



ZENER VOLTAGE REGULATOR

- The function of the zener diode is to maintain the output voltage more or less constant even if the load current changes.
- This is accomplished by operating the zener diode in the breakdown region when voltage across it changes only very slightly over a wide variation of zener current.
- The zener breakdown voltage has to be lower than the applied voltage, V.
- A zener diode will maintain a constant voltage across its terminals even if there is a change in the supply voltage, V in the circuit as shown in the figure.

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Bipolar Junction Transistor

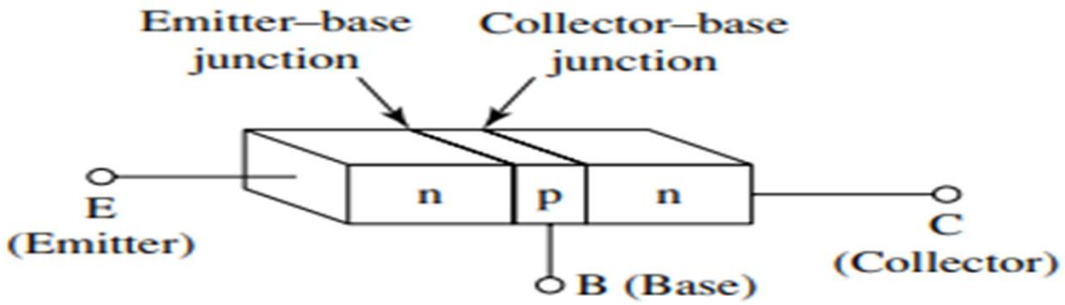
- A Bipolar Junction Transistor (BJT) is a three terminal semiconductor device in which the conduction depends on both majority and minority carriers and hence the name Bipolar.
- Dr. William Shockley and Dr. John Bardeen at Bell Laboratory in America invented the transistor in 1951. Since then, it has revolutionized the field of electronics.
- The BJT has replaced the bulky vacuum tubes and is comparatively smaller in size.
- It is used in amplifier and oscillator and as an electronic switch in digital circuits.
- It got wide applications in computers, satellites and communication systems.

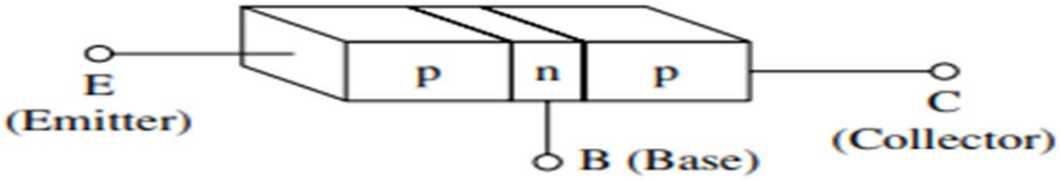
Construction of BJT

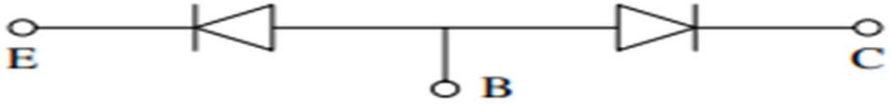
A BJT is a single crystal of silicon (Si) or germanium (Ge) that consists of three differently doped semiconductor regions, the emitter region, the base region, and the collector region. These regions are p-type, n-type, and p-type, respectively, in a PNP transistor, and n-type, p-type, and n-type in an NPN transistor. This can be illustrated as shown in figure (1) and treated as sandwiching p-type (N) between two n-type (P) semiconductor layers. Each semiconductor region is connected to a terminal, appropriately labelled as emitter (E), base (B), and collector (C), and the functioning of the three regions is as given below:

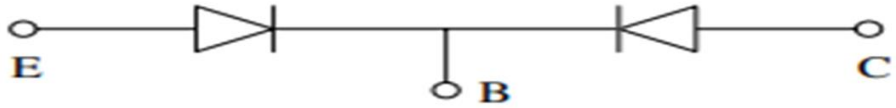
(i) Emitter (E): It is an outer region situated on one side of a transistor. The function of the emitter is to inject charge carriers like electrons in the case of an NPN transistor and holes in the case of a PNP transistor into the base. The emitter is heavily doped to supply many charge carriers.

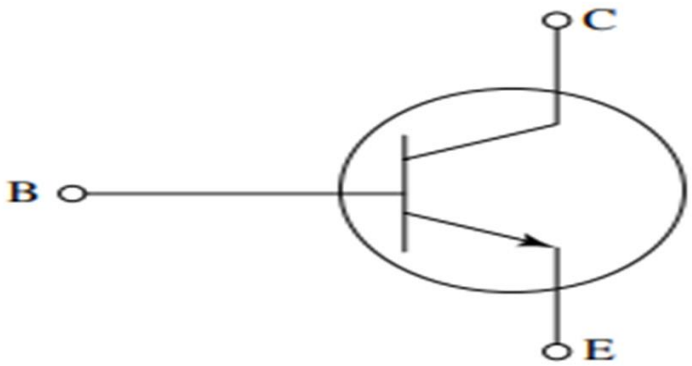
Construction of BJT



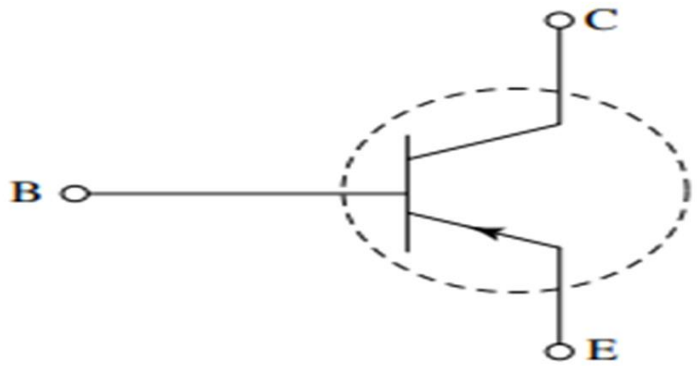








(a)



(b)

Construction of BJT

(ii) Base(B): The base is physically located between the emitter and the collector. It is generally a lightly doped, very thin layer. The function of this region is to pass all the charge carriers (electrons or holes) into the collector.

(iii) Collector (C): It is the outer region situated on the other side of the transistor. The collector's doping is located between the heavy doping of the emitter and the lightly doping of the base. The function of the collector is to collect charge carriers. The collector region is physically larger than the emitter region, as the collector has to dissipate more power.

At the outset, a transistor has two PN junctions. One junction is formed between the emitter and the base and is known as the emitter-base junction. Another junction is formed between the base and the collector and is known as the collector-base junction. The bipolar junction transistor is usually not a symmetrical device. This means that by swapping the collector and emitter, their functions cannot be interchanged.

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Construction of BJT

When the transistor is used as a circuit element in any electronic circuit, it is represented by the symbol as shown in figure . Figure (a) shows the symbol of the PNP transistor. The arrowhead in the emitter points from the p-region towards the n-region , indicating the Conventional current flow from the p-type emitter to the n-type base .

Figure shows the symbol for an NPN transistor. The arrowhead in the emitter points from the p-region towards the n-region, indicating the conventional current flow from the n-type emitter to the p-type base . The NPN transistor is the more common of the two types of transistors. The reason for this is that in NPN transistor,the current conduction is mainly due to electrons, whereas in PNP is that in the transistor conduction is due to holes. Since the electrons' mobility is higher than that of the holes, the conduction is higher in NPN transistors.

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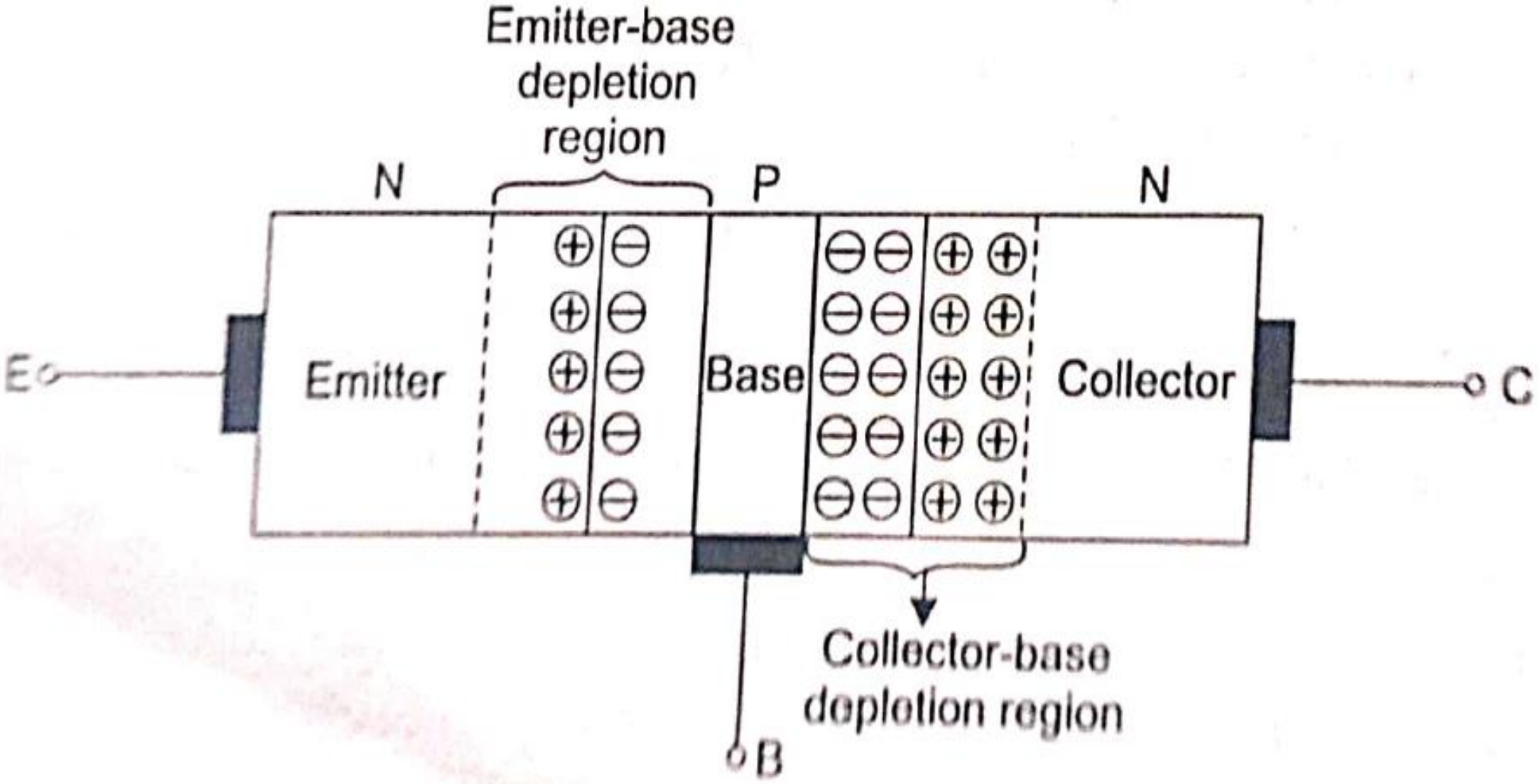
Working Principle of Transistor

A Bipolar Junction Transistor (BJT) is a semiconductor device with three layers of semiconductor material: the emitter, base, and collector. These layers are typically made of either silicon or germanium. The BJT operates based on the movement of charge carriers (electrons and holes) within these layers, and it comes in two major types: NPN (Negative-Positive-Negative) and PNP (Positive-Negative-Positive). The semiconductor layers of a BJT are intentionally doped to create regions with excess charge carriers. In an NPN transistor, the emitter is heavily doped with electrons (n-type), the base is lightly doped (p-type), and the collector is moderately doped (n-type). In a PNP transistor, the doping types are reversed. A BJT operates based on the control of current flow between the emitter and collector by manipulating the flow of charge carriers through the base region. This control enables amplification and switching functions, making BJTs a fundamental component in many electronic circuits.

Working Principle of Transistor

(i) Transistor with no bias: If no external DC voltage is applied between the different terminals of a transistor, then the transistor is said to be in an unbiased state or in an open- circuit state, as shown in figure . Consequently, the diffusion of charge carriers across the junction produces two depletion regions. The barrier potential for each of these depletion layers is 0.7 V for Si and 0.3 V for Ge. But the three regions have different doping levels, so the depletion widths are not the same for the two junctions. Because the emitter is more heavily doped than the base and collector, the width of the emitter-base depletion region is smaller than the width of the collector-base depletion region width

Transistor with no bias



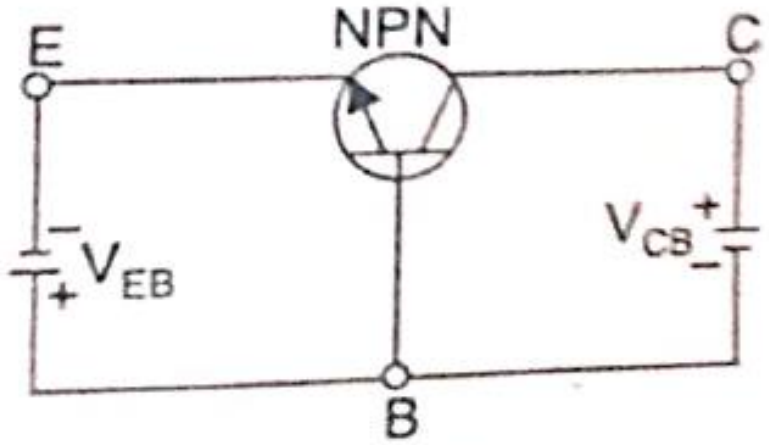
Working Principle of Transistor

(ii)Modes of Operation of Transistor: When dc voltages are applied across the different terminals of a transistor , then this process is called biasing. There are two junction transistor namely emitter-base junction and collector-base junction and each of these two junctions may be forward-biased or reverse-biased. So, there are four possible ways of beating these two junctions as shown in figure (7.4) and are called modes of operation of a transistor These are listed in table (7.1).

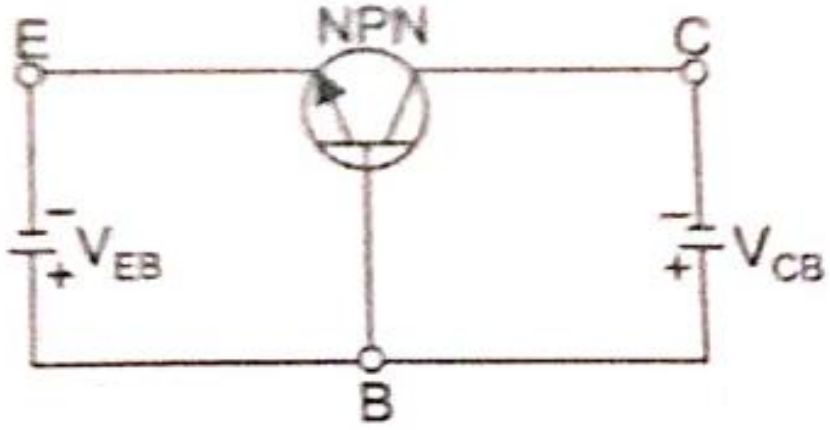
Modes of operation of BJT

- **Saturation** — The transistor acts like a **short circuit**. Current freely flows from collector to emitter.
- **Cut-off** — The transistor acts like an **open circuit**. No current flows from collector to emitter.
- **Active** — The current from collector to emitter is **proportional** to the current flowing into the base.
- **Reverse-Active** — Like active mode, the current is proportional to the base current, but it flows in reverse. Current flows from emitter to collector (not, exactly, the purpose transistors were designed for).

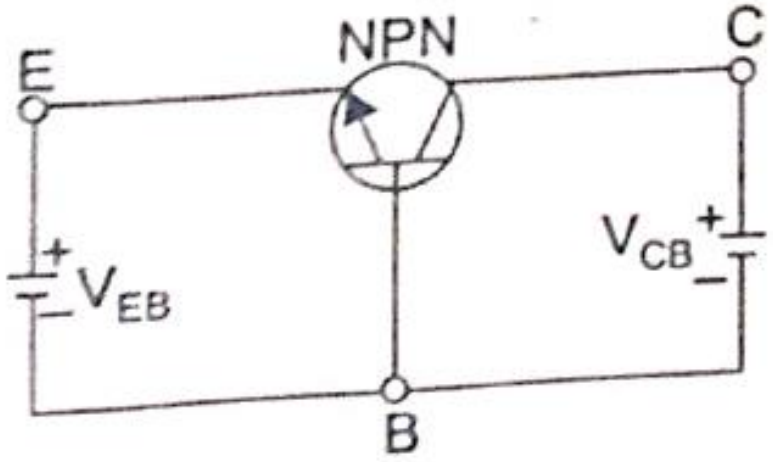
Modes of operation of BJT



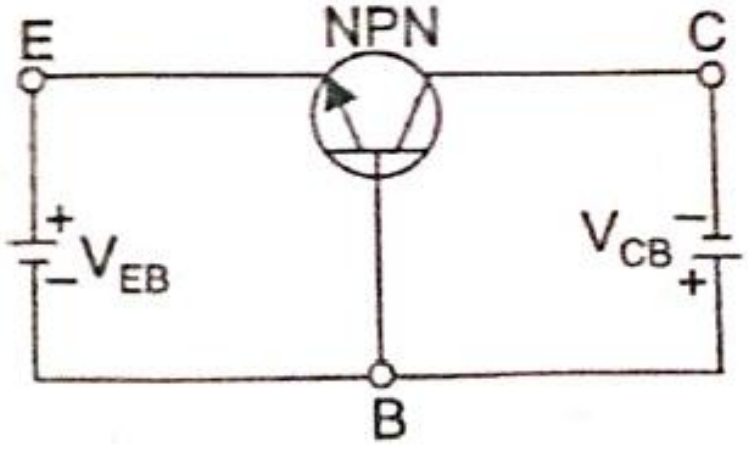
(a) Active Region



(b) Saturation Region



(c) Cut off Region



(d) Inverted Region

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Modes of operation of BJT

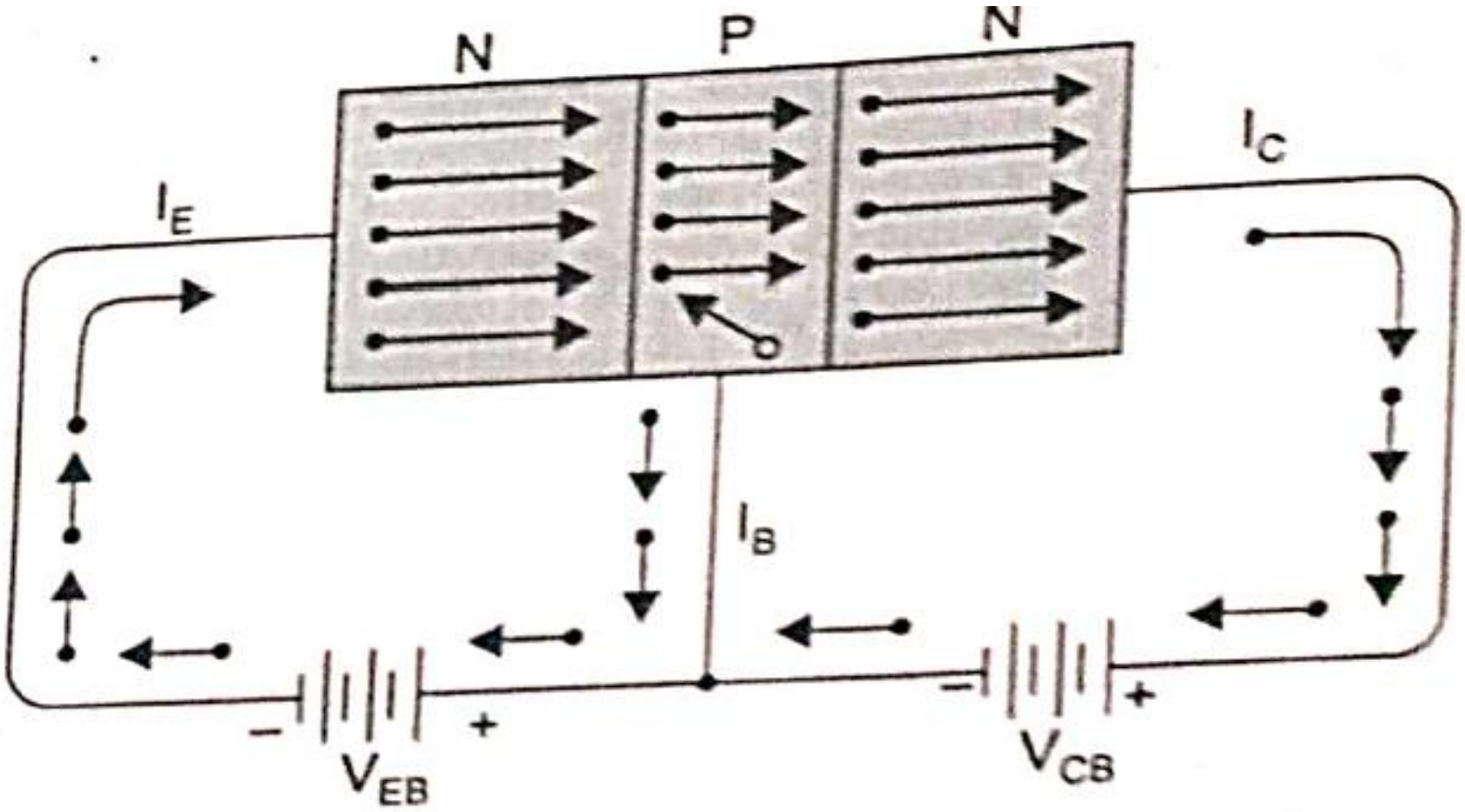
Table (7.1): *Modes of operation of BJT*

Mode	Emitter-base junction	Collector-base junction	Region of operation
I	Forward biased	Reverse biased	Active region
II	Forward biased	Forward biased	Saturation region
III	Reverse biased	Reverse biased	Cut-off region
IV	Reverse biased	Forward biased	Inverse active region

Operation of NPN transistor:

Consider an NPN transistor that biased to operate in the active region, as shown in figure (7.5). The emitter-base junction is forward-biased ,and the collector-base junction is reverse-biased. The forward bias applied to the emitter base junction of an NPN transistor allows many electrons to flow from the emitter region towards the base region. As well, a few holes also move from the base region to the emitter region. This flow of electrons and holes forms the emitter current I_E .As the base is lightly doped with p-type impurities, the number of holes in the base region very small. Hence, a few electrons combine with the holes to constitute a base current I_B .The remaining electrons cross into the collector region to constitute a collector I_C .Thus, the base and collector currents summed up give the emitter current, i.e.. $I_E = I_B + I_C$

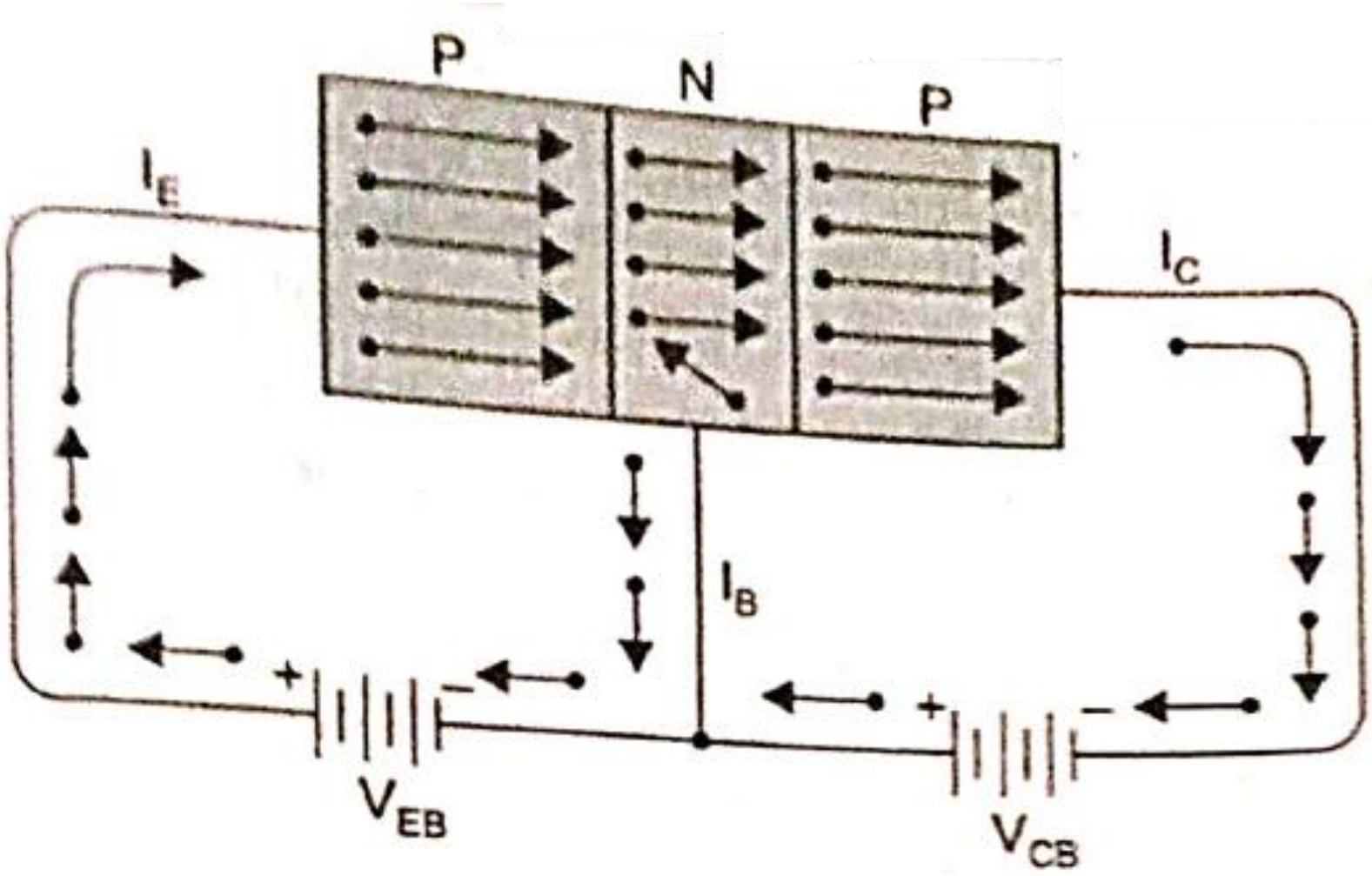
Operation of NPN transistor:



Operation of PNP transistor:

As shown in figure (7.6), a PNP transistor is biased to operate in the active region. The forward bias is applied to the emitter-base junction of a PNP transistor. This forward bias injects many holes from the emitter region to cross the base region as the base is lightly doped with N-type impurities. The electrons in the base region are minority carriers and are very few in number. Hence, the number of holes recombined with electrons in the N-type base region is also small. Hence, a few holes combine with electrons to constitute a base current I_B .The remaining holes (approximately 95% of the total) move into the collector region to form collector current I_C .Hence, the emitter current is the sum of the collector current and the base current. The three currents, namely the emitter current, base current, and collector current, are connected in the PNP transistor's external circuit by, $I_E = I_B + I_C$

Operation of PNP transistor



TRANSISTOR CONFIGURATION

- When a bipolar junction transistor is connected in a circuit, one terminal is used as an input terminal, the second terminal is used as an output terminal, and the third terminal is common to the input and output. Depending upon these terminals, a transistor can be connected in three configurations, such as the

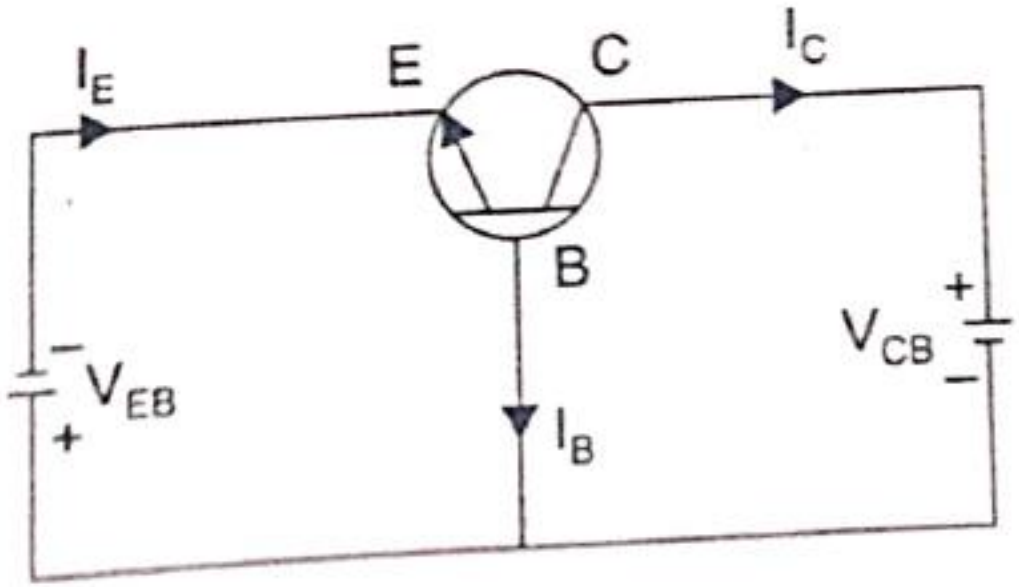
(i) common base (CB) configuration,

(ii) common emitter (CE) configuration, and

(iii) common collector (CC) configuration.

Common Base Configuration

In a common-base transistor configuration, the emitter is the input terminal, the collector is the output terminal, and the base of the transistor is common to both terminals. This is also called the grounded base configuration. The circuit diagram of the common-base configuration is shown in figure (7.7). The behavior of the transistor is analyzed with the help of its input and output characteristics.



Common Base Configuration

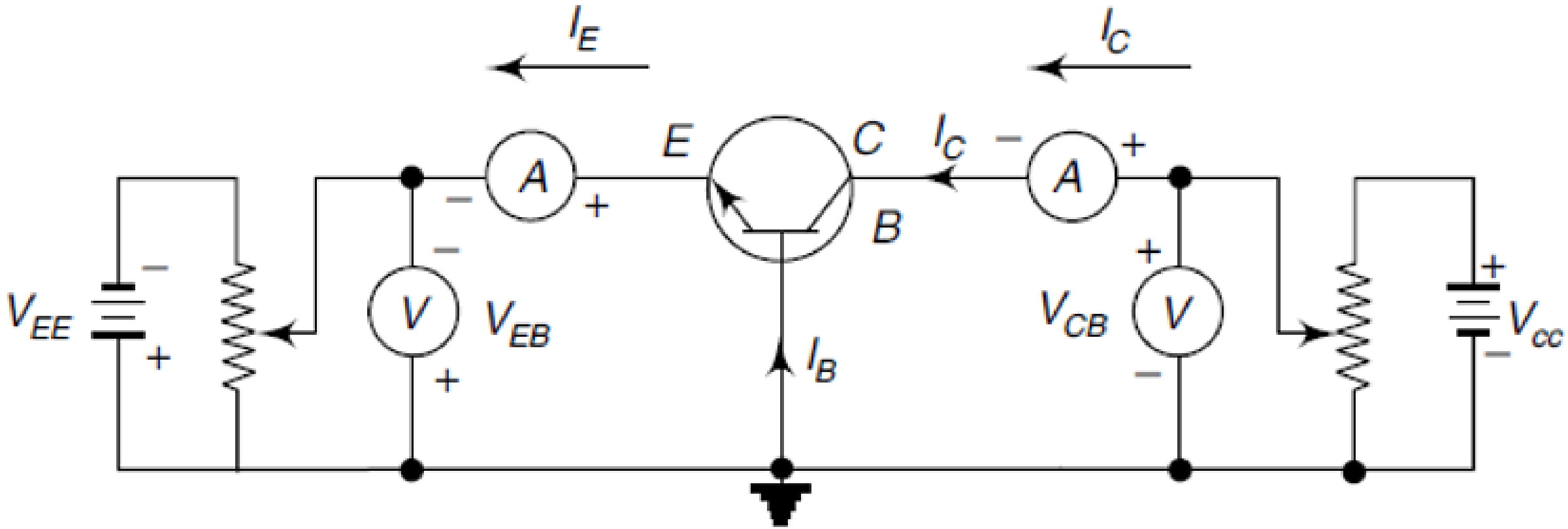


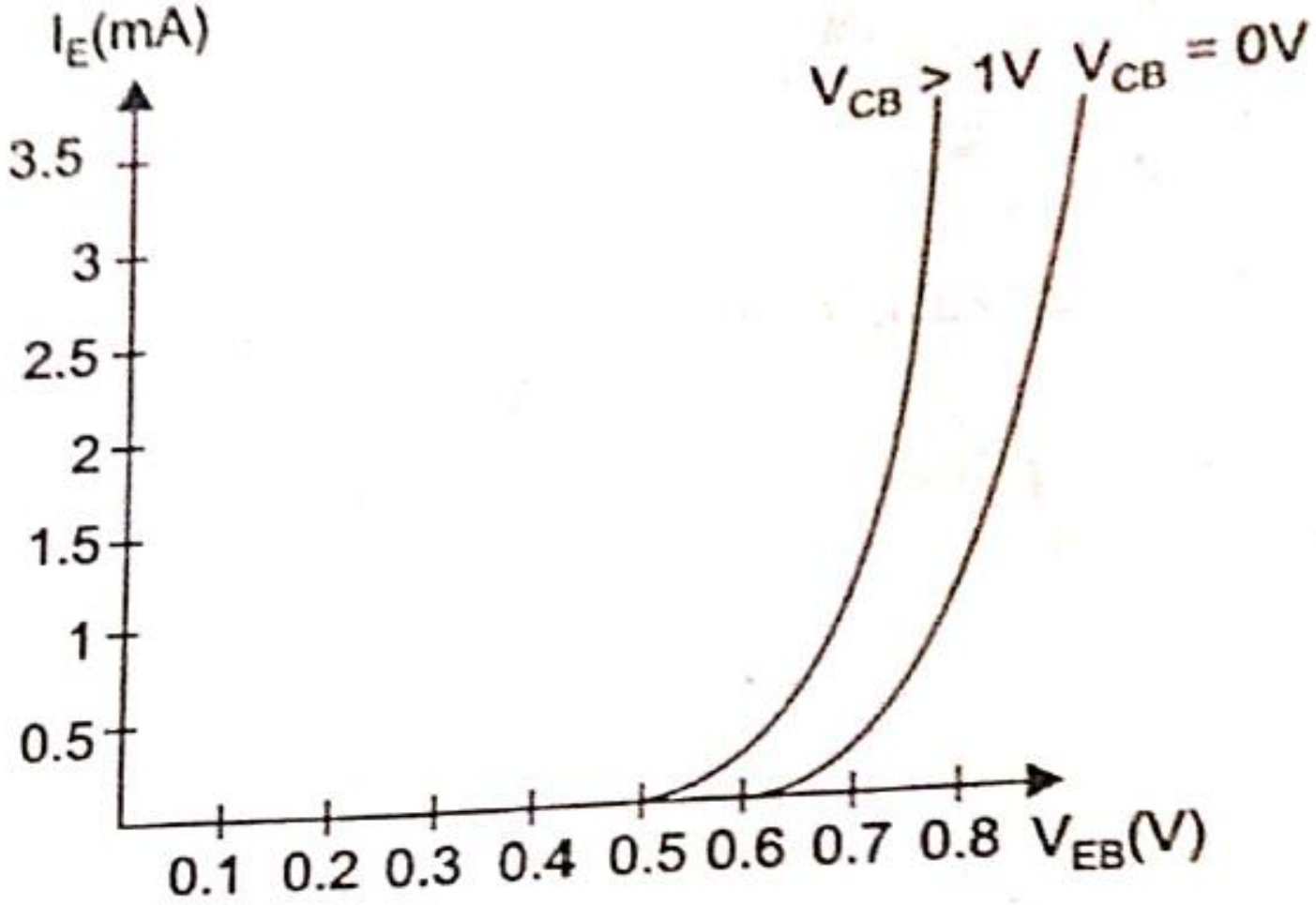
Fig. *Circuit to determine Common-base characteristics*

Common Base Configuration

Input Characteristics: The input characteristics curves defines the relation between the input or emitter current I_E and emitter to base voltage V_{EB} at a constant output voltage V_{CB} The input characteristics of the transistor are determined by keeping the output voltage collector –base voltage V_{CB} is kept constant at zero volt and the emitter current IE is increased from zero in suitable steps by increasing V_{EB}. When V_{CB} is equal to zero, the emitter base junction is forward biased and the junction behaves as a forward biased diode. The emitter current I_E increases rapidly with small increase in emitter base voltage V_{EB}. When V_{CB} is increased keeping V_{EB} constant, the width of the base region will decrease. This results an increase in I_E The common Base input characteristics are shown in figure (7.8).

Common Base Configuration

Input Characteristics:

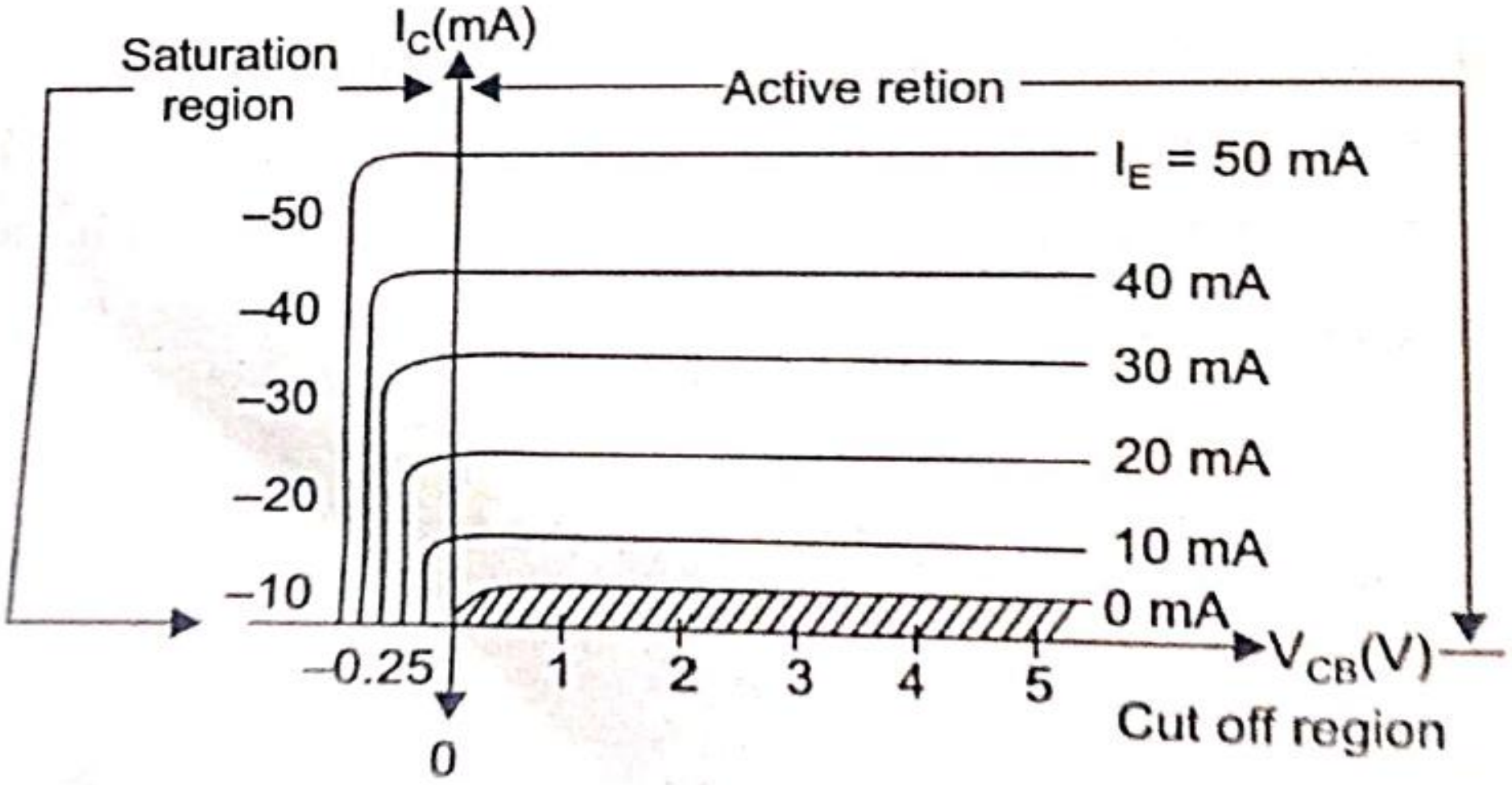


Common Base Configuration

Output Characteristics: The output characteristic curves relate the output or collector current I_C and collector to base voltage V_{CB} keeping the input or emitter current I_E constant. To determine the output characteristics, emitter current I_E is kept constant. Then V_{CB} is increased in uniform steps and the collector I_C is noted for each value of I_E. This is repeated for different values of I_E and is plotted as shown in figure (7.9). The characteristic curves shows that at a constant value of I_E,I_C is independent of V_{CB}. Moreover, I_C flows even when V_{CB} is zero.

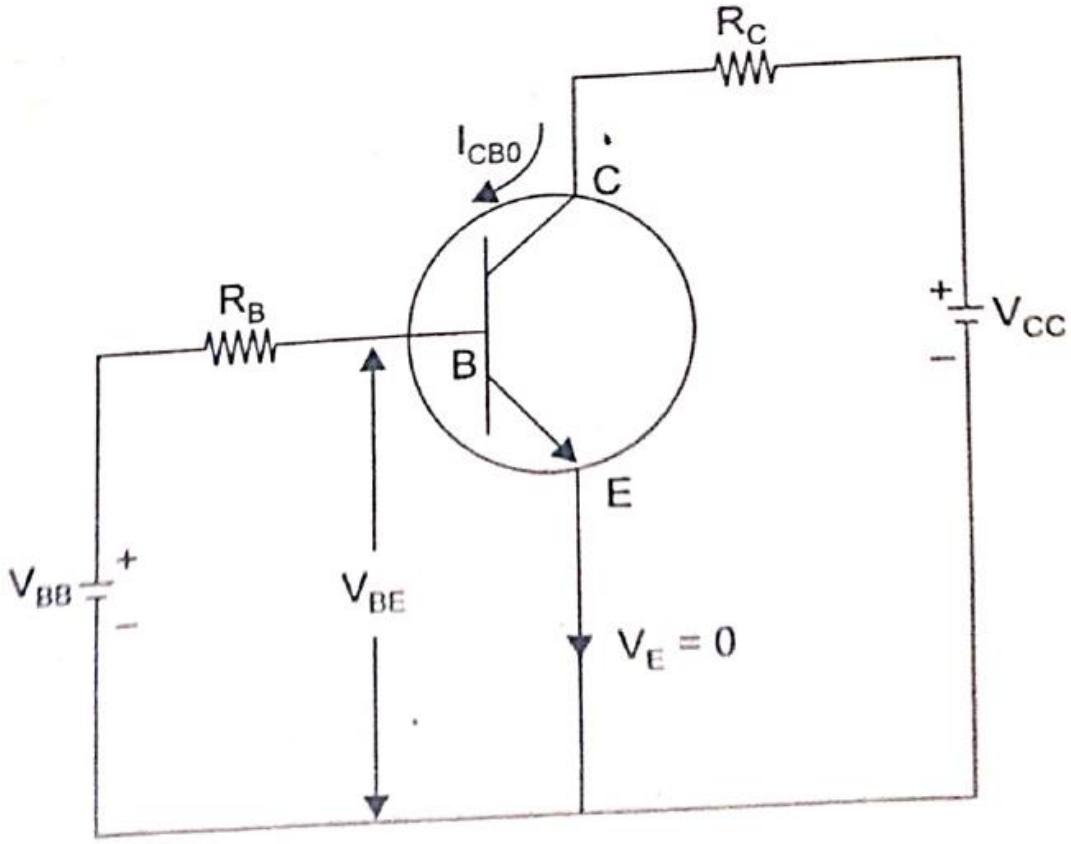
Common Base Configuration

Output Characteristics



Common Emitter Configuration

In common emitter configuration, base is the input terminal, collector is the output terminal and emitter is common to both the terminals. This is also called as grounded emitter configuration. The circuit of common-emitter configuration is shown in figure .The behavior of the transistor is analyzed with the help of the input characteristics and output characteristics.



Common Emitter Configuration

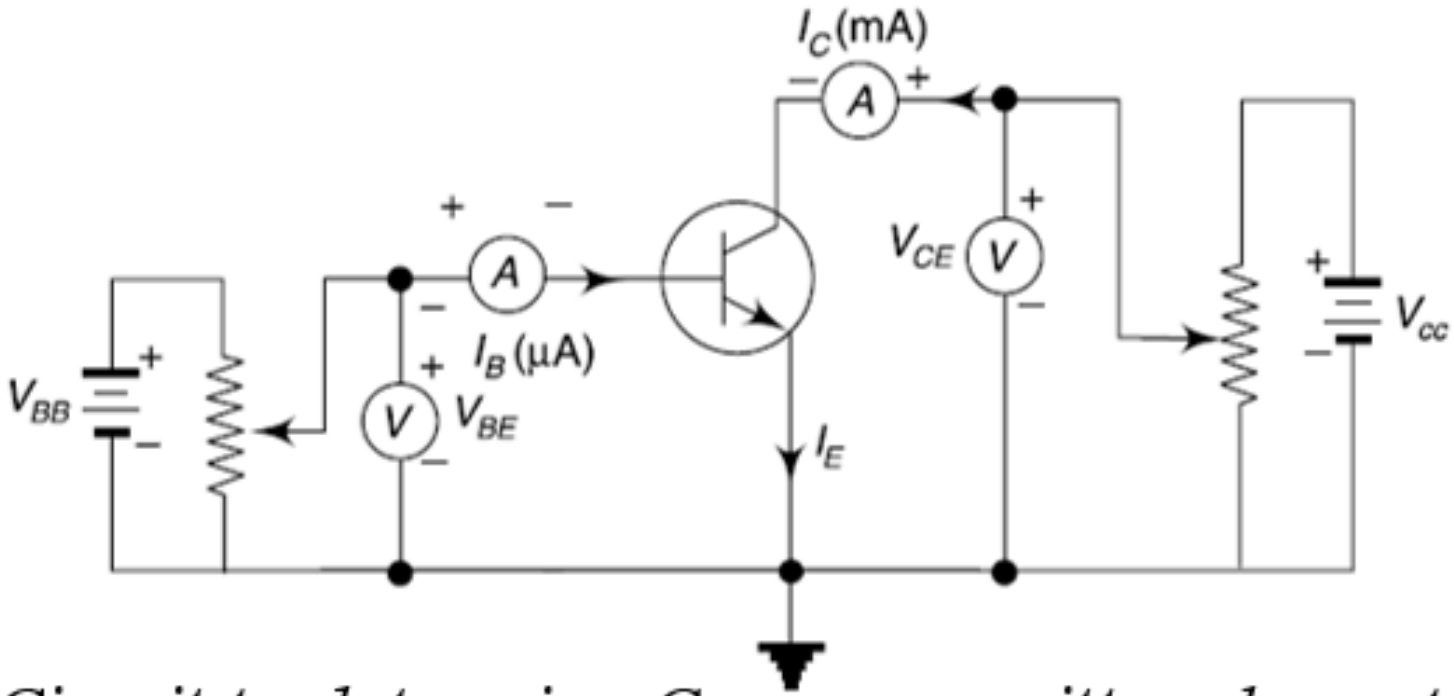


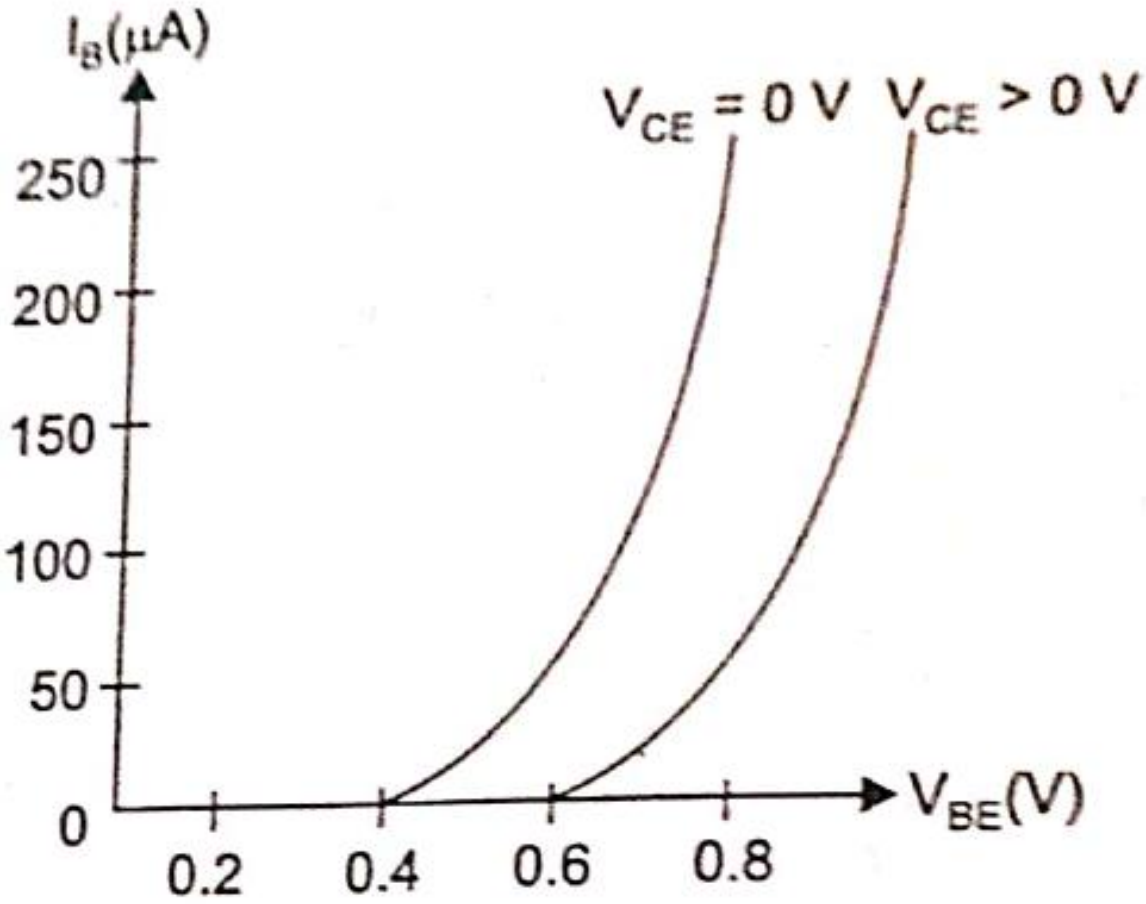
Fig. *Circuit to determine Common-emitter characteristics*

Common Emitter Configuration

Input characteristics: The input characteristic curve defines the relation the base current I_B and the base to emitter voltage V_{BE} at a constant voltage V_{CE}. The input characteristics are obtained by increasing the base current in equal steps by increasing V_{BE} a fixed value of V_{CE}, zero voltage. When V_{CE} is zero, the emitter base junction is forward biased and the behaves as a forward biased diode. So, the input characteristics resemble the forward biased diode at V_{CE} =0. When V_{CE} is increased, the width of the depletion region at the reverse biased collector base junction will increase. This will reduce the effective width of the base and leads to decrease in the base current I_B. In order to provide the same amount of I_B,V_{BE} should be increased. Hence the curves appear to the right as V_{CE} increases. The input characteristics of common emitter configuration are shown in figure (7.11).

Common Emitter Configuration

Input characteristics:

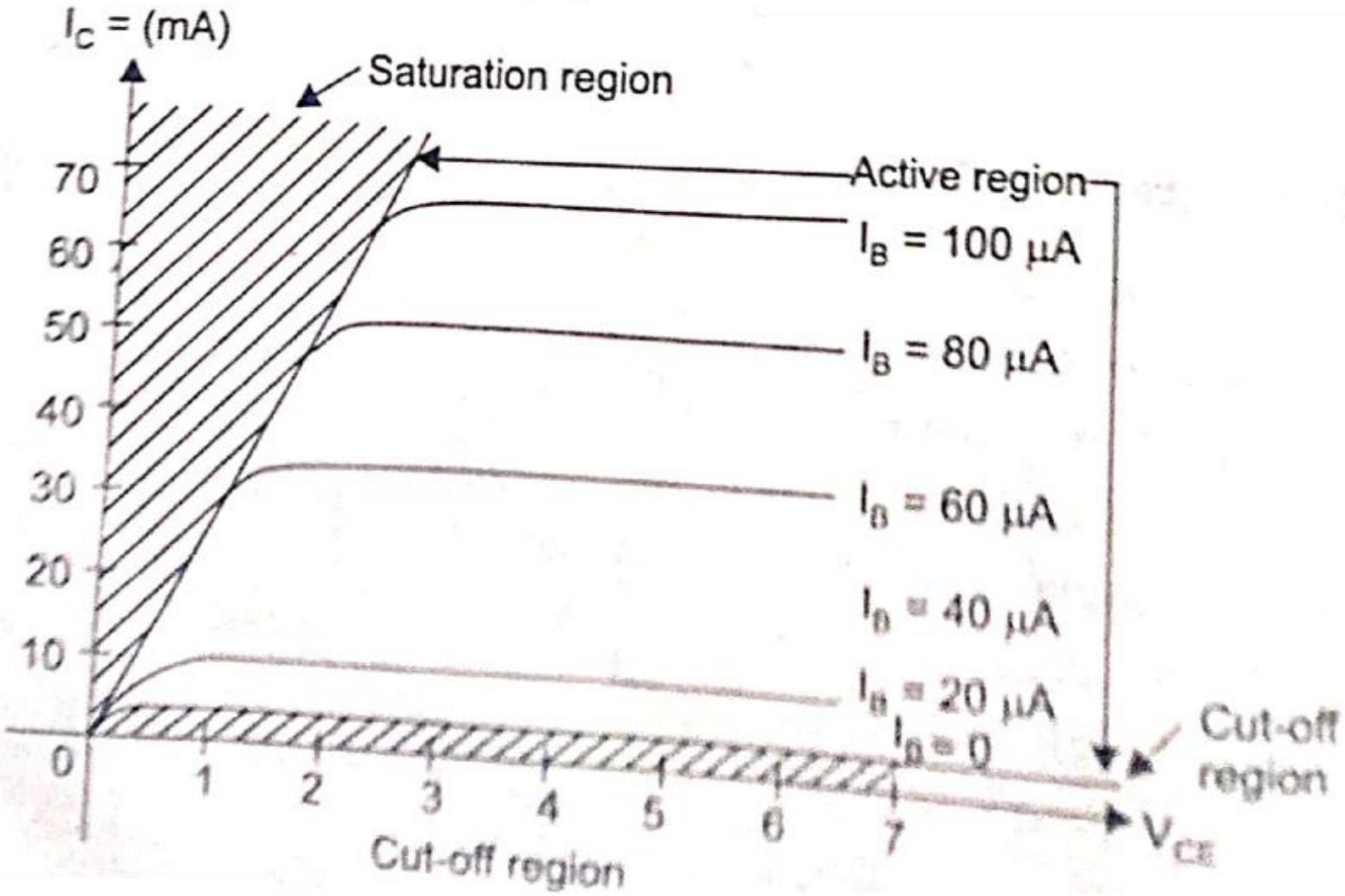


Common Emitter Configuration

Output Characteristics: The output characteristic curves defines the relation between the collector current I_C and collector to emitter voltage V_{CE} at a constant base current I_B. The family of curves is shown in figure (7.12). The output characteristics are obtained by the change in collector current I_C due to a change in V_{CE} at constant input current I_B. As we change the output voltage V_{CE} and the input voltage overline V_{BE}, this will affect the biasing of the both the junctions and in turn the operating regions of the transistor. Hence, the output characteristics have three regions, namely, active region, saturation region and cut-off region.

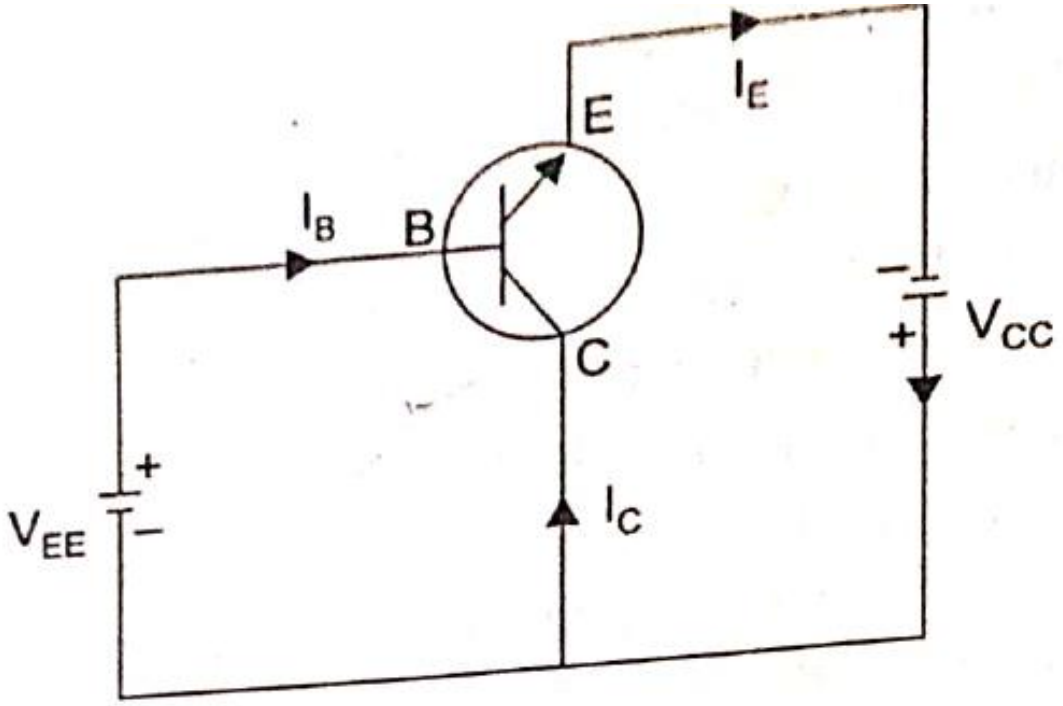
Common Emitter Configuration

Output Characteristics:



Common Collector Configuration

In common collector configuration, the base is the input terminal, the emitter is the output terminal and collector is the common terminal. This is known as grounded collector configuration. The circuit for obtaining the characteristics of an NPN transistor in CC configuration is shown in figure (7.13).



Common Collector Configuration

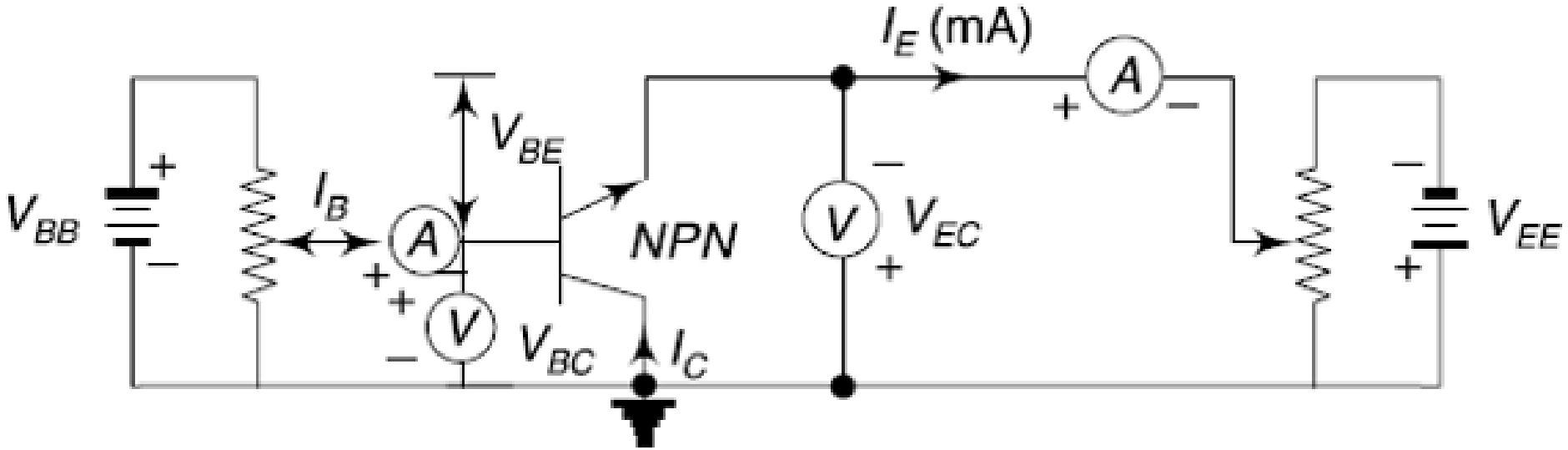
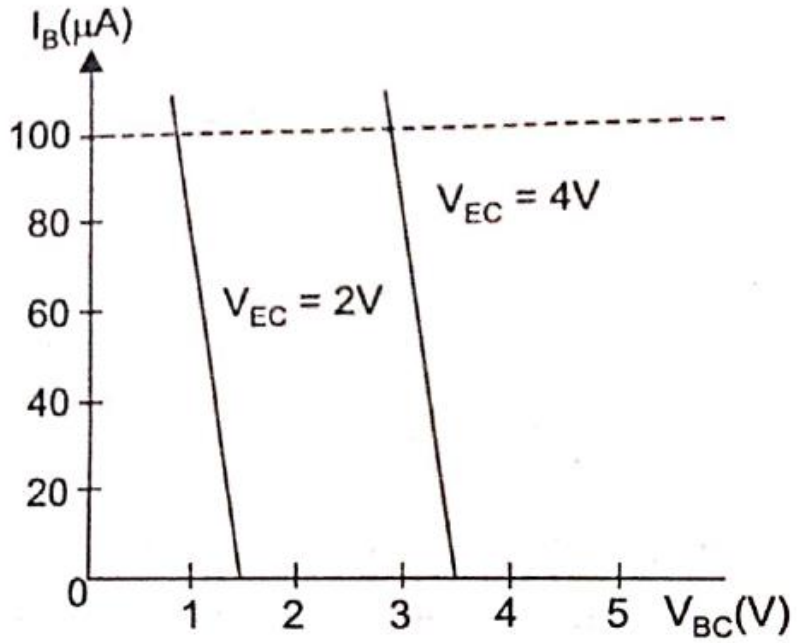


Fig. *Circuit to determine Common-collector characteristics*

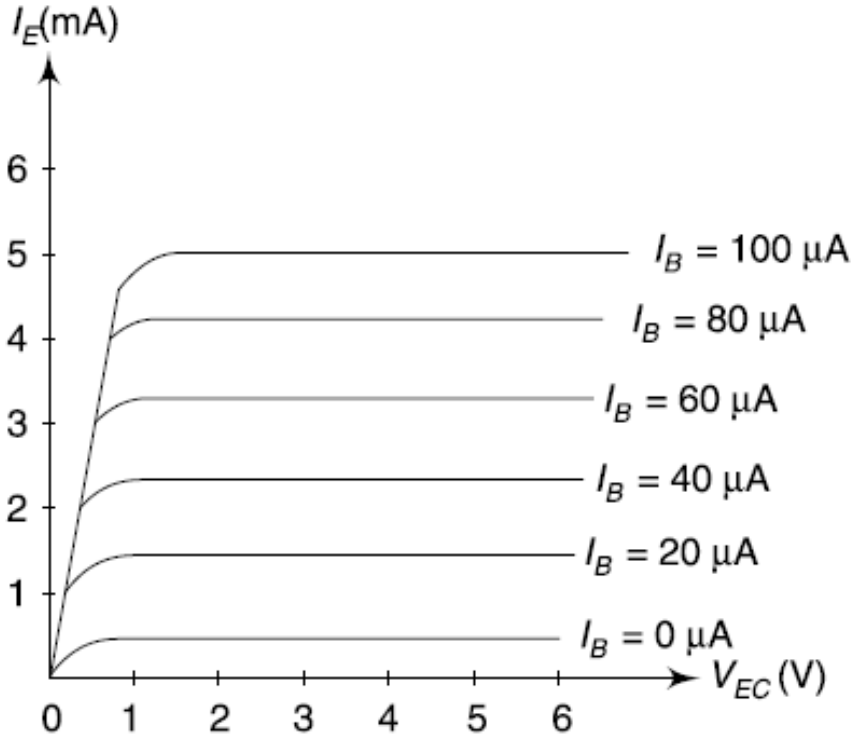
Common Collector Configuration

Input Characteristics: The input characteristic curves define the relation between the base current I_B and the base to collector voltage V_{BC} at a constant voltage V_{EC} .The input characteristics are obtained at a fixed voltage of V_{EC} by increasing the base collector voltage V_{BC} in uniform steps and the corresponding increase in I_B is noted. This process is repeated different values of V_{EC}. The plot of input characteristics curves is shown in figure



Common Collector Configuration

Output Characteristics: The output characteristic curves define the relation between the emitter current I_E and emitter to collector voltage V_{EC} at a constant base current I_B. The family of characteristic curves is similar to the common emitter configuration as the output current, i.e. emitter current is almost equal to the collector current. The output characteristics of CC configuration are shown in figure (7.15).



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Comparison between CB,CE and CC

Property	*CB*	*CE*	*CC*
Input resistance	Low (about 100 Ω)	Moderate (about 750 Ω)	High (about 750 kΩ)
Output resistance	High (about 450 kΩ)	Moderate (about 45 kΩ)	Low (about 25 Ω)
Current gain	1	High	High
Voltage gain	About 150	About 500	Less than 1
Phase shift between input & output voltages	0 or 360°	180°	0 or 360°
Applications	for high frequency circuits	for audio frequency circuits	for impedance matching

Transistor as an amplifier

A load resistor R_L is connected in series with the collector supply voltage V_{CC} of CB transistor configuration as shown in Fig. A small change in the input voltage between emitter and base, say ΔV_i, causes a relatively larger change in emitter current, say ΔI_E. A fraction of this change in current is collected and passed through R_L and is denoted by symbol α'. Therefore the corresponding change in voltage across the load resistor RL due to this current is $\Delta_{V0} = \alpha' R_L \Delta I_E$.

Here, the voltage amplification $A_v = \frac{\Delta V_0}{\Delta V_i}$ is greater than unity and thus the transistor acts as an amplifier.

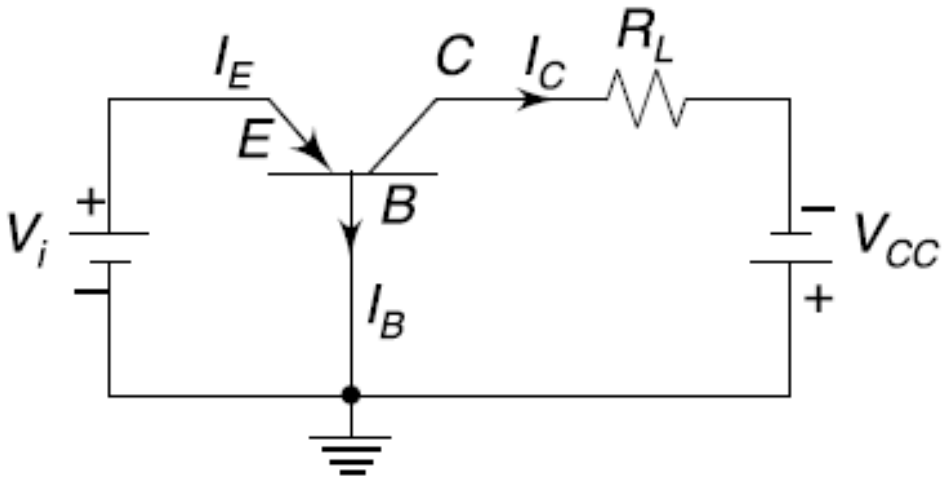


Fig. *Common base transistor configuration*

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Thank You